OPENWINDOW V1 · BUILD AND OWNER'S GUIDE

OpenWindow — a 3D-printable, double-glazed awning window

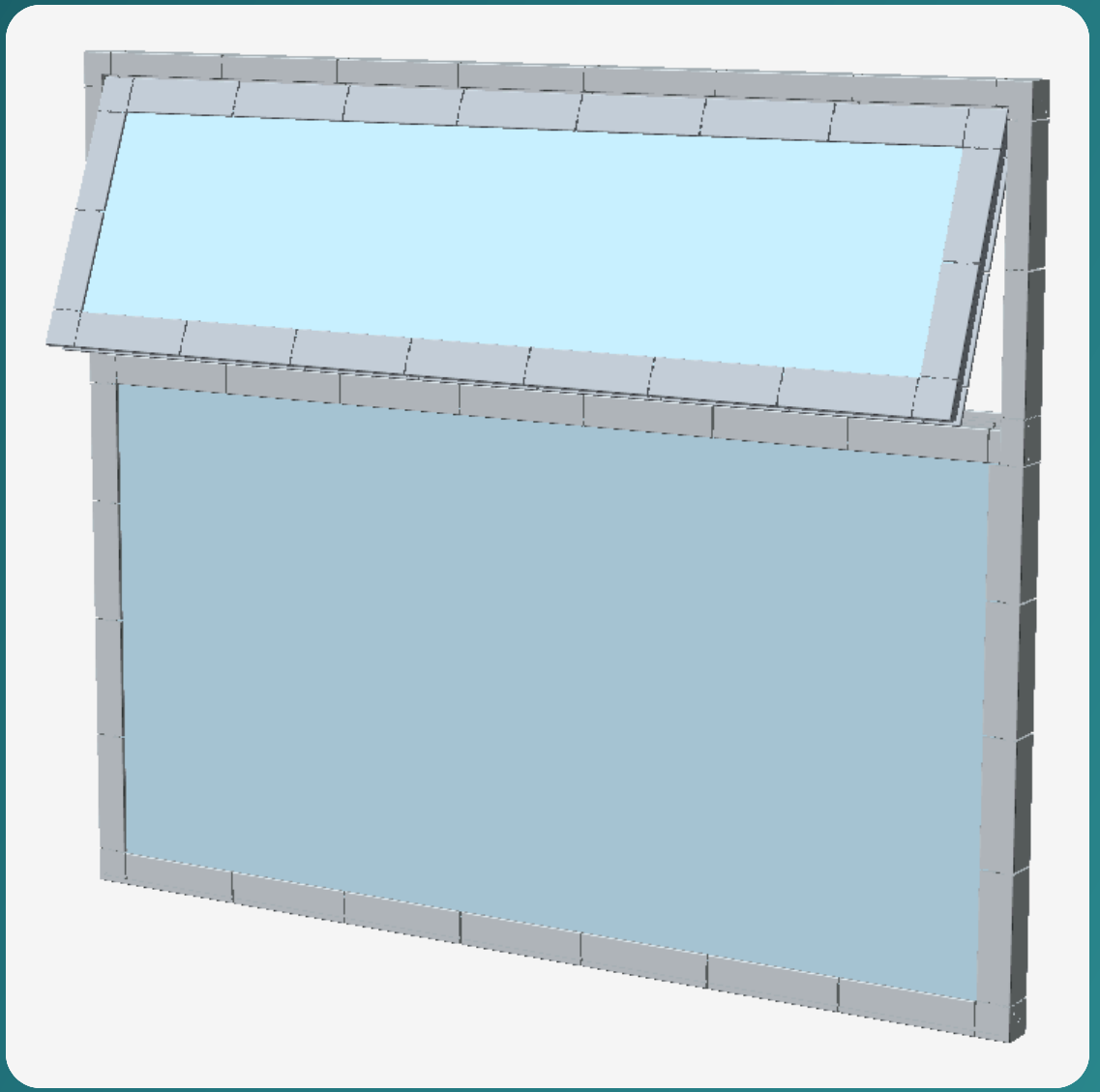
Size it for your opening, print it on a Bambu P1S, assemble it on the bench, fit it into a brick-veneer wall, and look after it: the whole job, step by step.

1800 × 1500 mm
home window: top awning over a fixed IGU

160 prints
35 part types, 16.36 + 3.84 kg filament

63 mm deep
printed shells on aluminium and steel tubes

Bench window first
600 × 600 mm test build: 98 prints



Version 1 · 11 October 2026

Open source · licence: to be chosen

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Why OpenWindow, and what v1 is

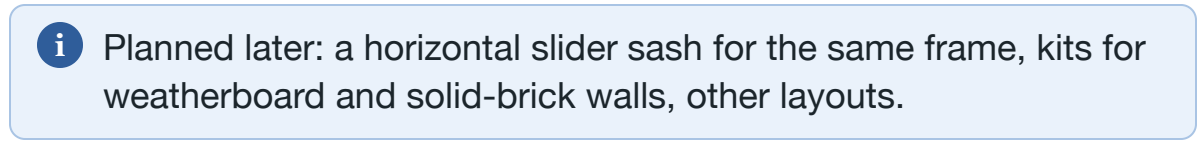
The problem

- Basic single-glazed aluminium sliders lose heat fast: about **6 W/m²K** through the glass.
- Brush-pile seals leak air, so they're draughty in wind and let insects in.

What OpenWindow aims for

- Double glazing: about **2.7–3.0 W/m²K** at the glass.
- **Airtight when closed:** EPDM compression seals, not brush pile.
- An insect screen on the room side.
- **Open source and parametric:** one small params file per opening; the generator does the rest.

v1 decision	
Replacement	Full: the old window comes out down to the wall opening
Wall	Brick veneer (timber frame + brick skin), typical of Melbourne
Opening	1500–1800 mm wide × 1200–1500 mm high
Layout	Top awning strip (about 450 mm high) over one fixed pane
Operation	Top-hung on stainless friction stays, two printed cam locks
Glazing	Universal 24 mm pocket: IGU, acrylic sandwich or twin-wall
Screen	Room side, hinged at the top, magnetic seal



The home window at a glance (1800 × 1500)

1784 × 1484
frame (mm) for a 1800 × 1500 opening, 63 deep

160 prints
35 part types, every one ≤ 240 mm long

16.36 + 3.84 kg
exterior filament (tier 3) + interior PETG

1698 × 878
fixed IGU (mm); sash IGU 1606 × 388

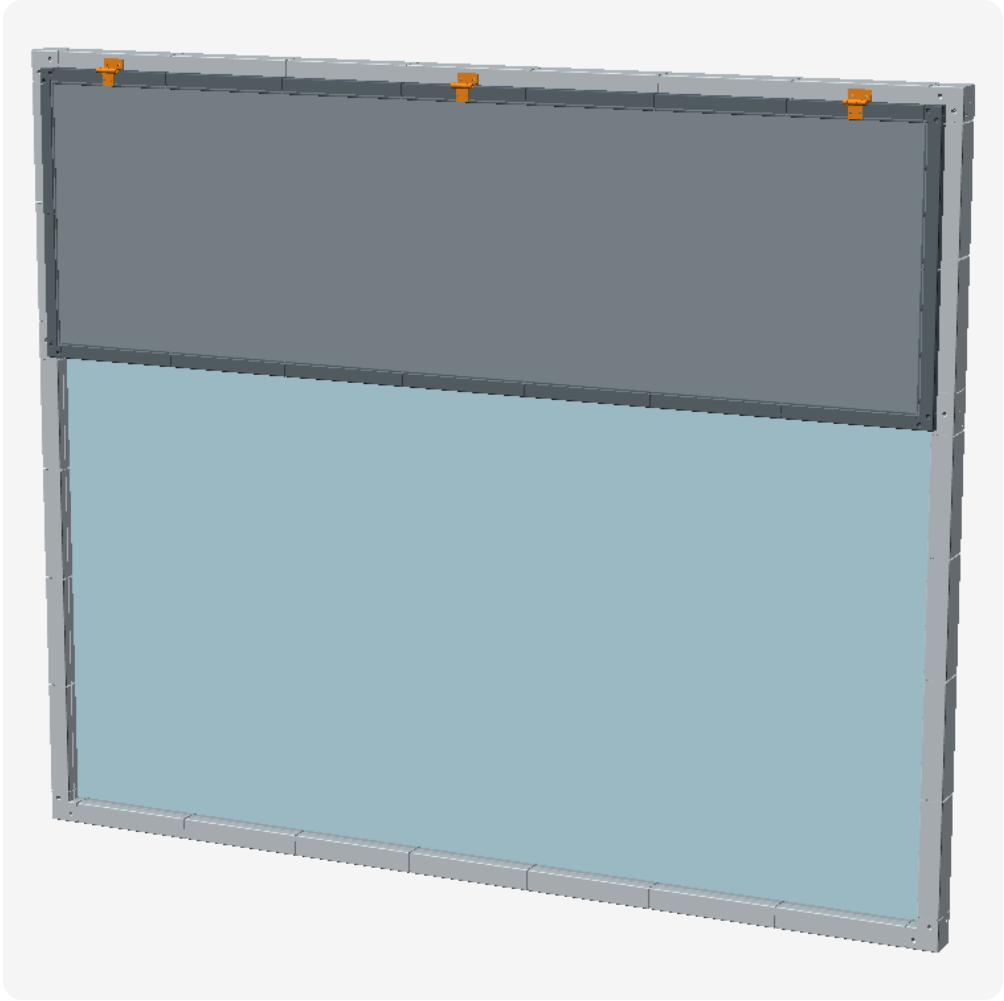
15.7 kg sash
on a 400 mm friction-stay pair

38 fixings
10g × 75 into the reveal, 44 packers

8.92 m
EPDM bulb seal in two loops round the sash

3 cartridges
±50 % MS-polymer sealant for joints and perimeter

Tubes: 10.31 m 20 × 20 alu, 4.37 m 10 × 10 alu, 1.7 m 40 × 20 steel.
Transom: deflects 6.2 mm at 0.7 kPa, limit span/175 = 9.5 mm.



Room side, screen closed. Every number on this slide comes from `bom/home-1800x1500.*`.

Honest limits (1): the size of the job

160
prints for the home window

20.2 kg
filament in total

≈ 550 h
printer time, spec estimate (§7)

2 people
to install; a dry day

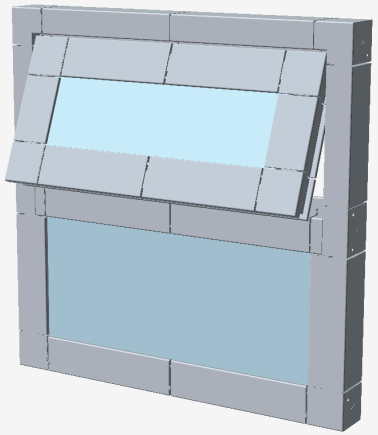
- Each printed segment is at most 240 mm long to fit the P1S bed, so a 1.8 m member has seven of them, and every joint is sealed by hand.
- The fixed IGU alone weighs about **30 kg**.
- Besides filament you buy metal tube, a steel transom tube, glass from a glazier, EPDM seals, sealant and stainless hardware.

Start small

The 600 × 600 mm **bench window** uses the same profiles, joints, seals and hardware:

- 98 prints, 6.19 + 1.25 kg filament;
- acrylic glazing, tier 1 PETG;
- mounted in a timber test box and put through the test protocol.

✓ Print and pass the bench window before you commit to the full size.



Honest limits (2): not certified, not yet proven

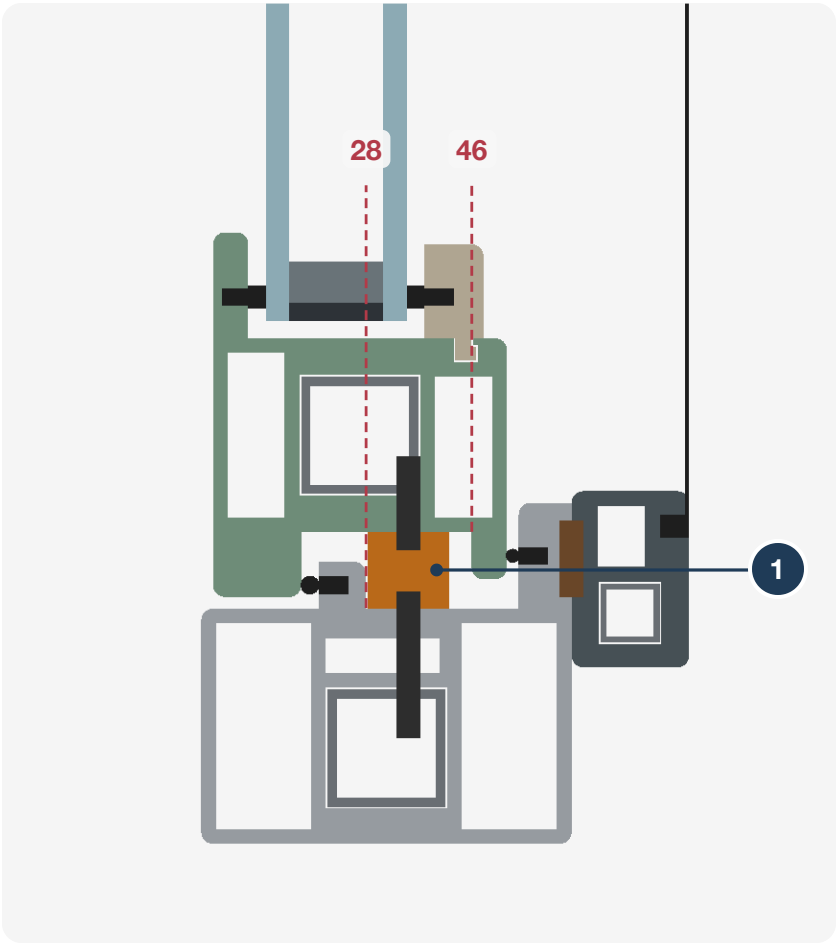
Not certified

- No **AS 2047** window rating (wind, water, air): nothing has been tested by a lab.
- Glass must follow **AS 1288**: toughened where it requires; buy it from a glazier.
- Not designed for bushfire-prone (**BAL**-rated) sites.
- Fit opening restrictors where the **NCC** requires them.
- Check your local building rules before you start.

Not yet proven

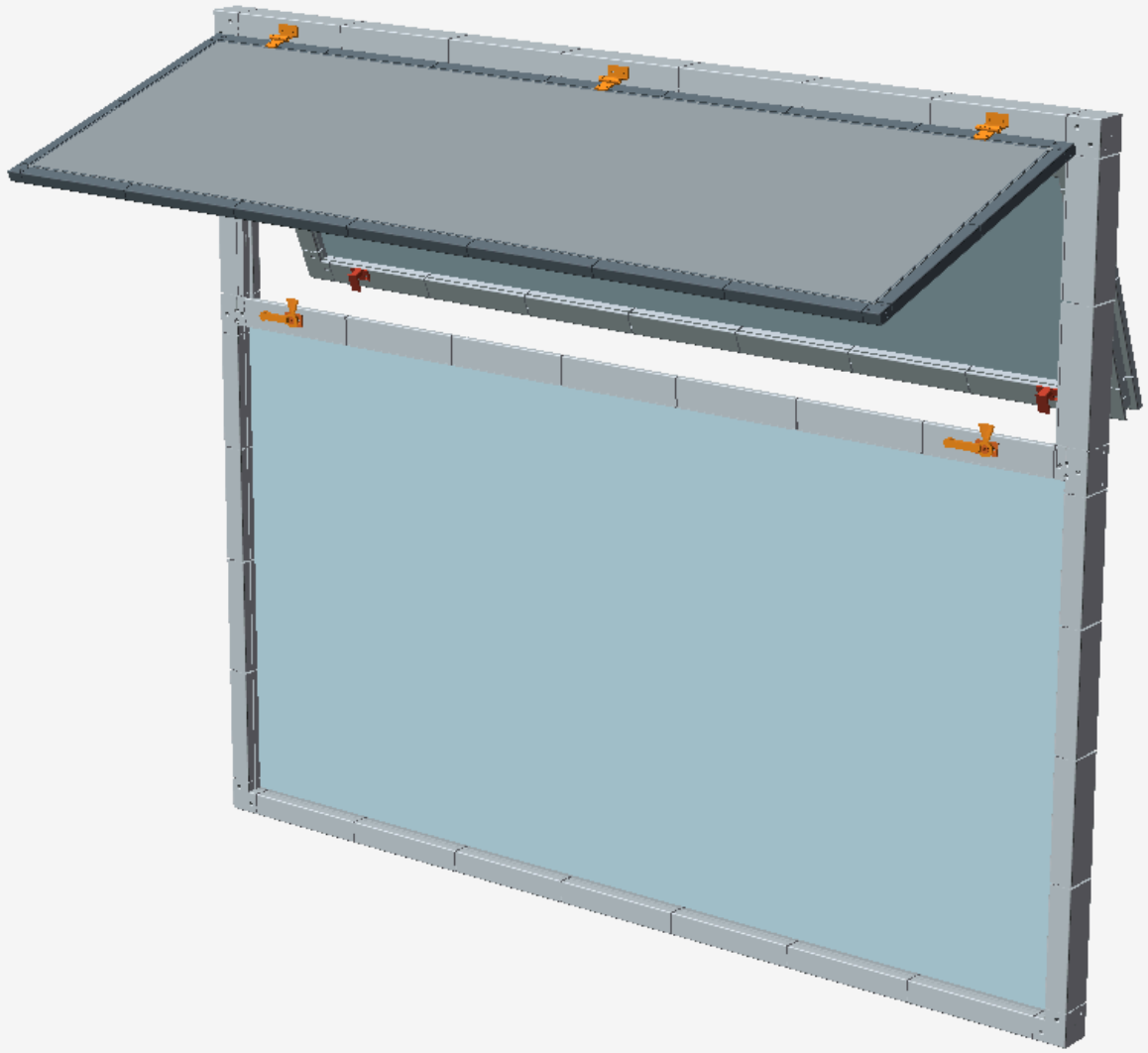
- **Friction-stay gate:** a 400 mm stay pair must fit a stack of ≤ 13 mm and a width of ≤ 14 mm, rated ≥ 15.7 kg. Many are 16–19 mm wide. If none fits, the CAD has to change.
- Joint sealing at the 1 mm expansion gap relies on a ±50 % sealant.
- Creep of printed parts in full sun, stays and corners under load, cam wear: all in the test protocol.

! Build and test the bench window first (Testing section). Order glass only after measuring the real opening.



The stay (1) must fold into the 13 mm gap between the jamb fin top (28) and the sash lip (46).

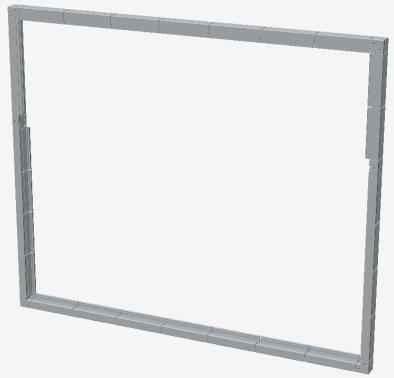
Layout: a top awning over a fixed pane



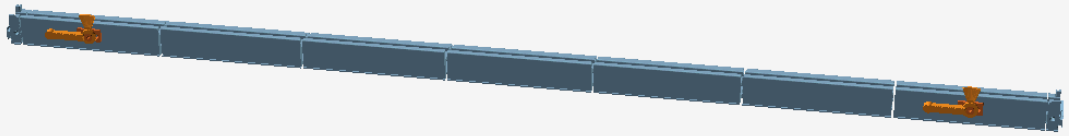
Seen from the room with the screen lifted and the sash pushed out.

- **Frame:** head, sill and two jambs, screwed to the timber reveal.
- **Transom:** a separate steel-cored bar. Its shoes slide down dovetails in the two jamb nodes after the frame is fixed.
- **Sash:** top-hung on a pair of stainless friction stays; two printed cam locks pull it onto its seals.
- **Fixed pane:** sits in the lower frame on printed setting blocks.
- **Screen:** hinged along the head, held shut by magnets; the cams sit inside it.

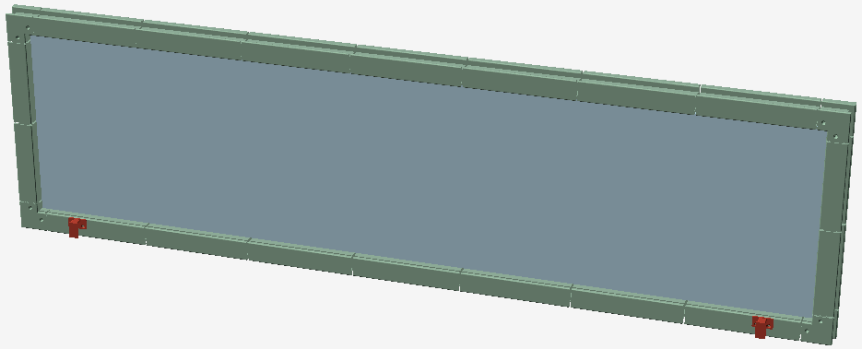
Four units, built flat on the bench



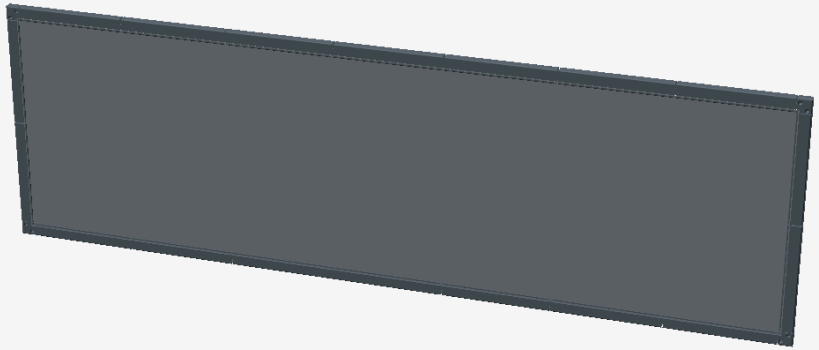
Frame 1784 × 1484 mm: sill, jambs with nodes, head, four corners



Transom with shoes and cam bases; dry-fitted at the bench, sealed at install



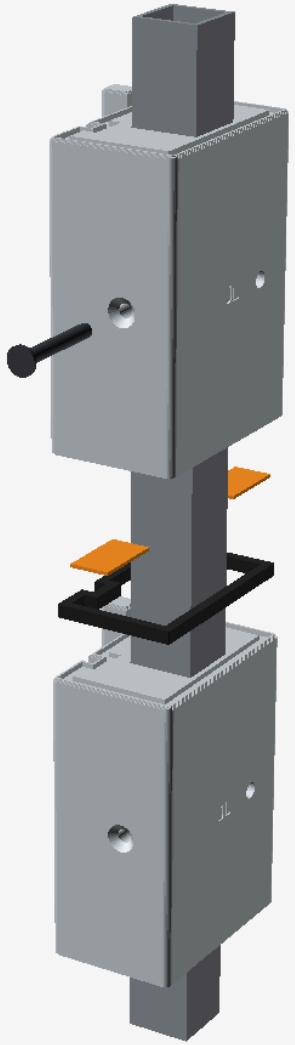
Sash 1678 × 460 mm, glazed at the bench, with keepers and stay arms



Screen 1724 × 540 mm with mesh and magnetic strip A

Plus the loose glass: the fixed 1698 × 878 mm IGU goes in after the frame is fixed (install step 9).

Structure: printed shells on continuous metal tubes

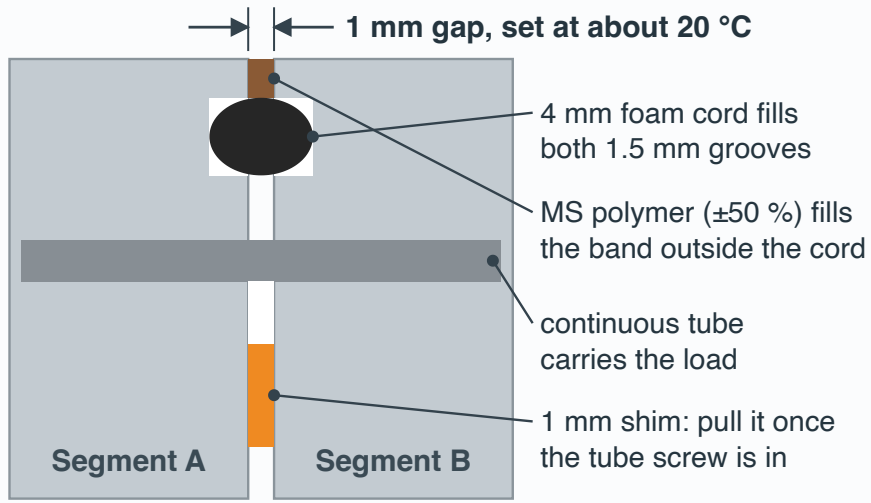


- Every member has **one continuous tube** running its full length.
- Printed segments, each **≤ 240 mm**, slide over it. One stainless screw through the inside face fixes each segment to the tube at its middle.
- **Corner blocks** take the tube ends in sockets; each tube stops 1.5 mm short of the socket floor.
- The tube carries the load. The joint between segments only aligns and seals.

Member	Core	Why
Head, sill, jambs	20 × 20 × 1.6 aluminium	screwed to the reveal every 400 mm
Transom	40 × 20 × 1.6 galvanised steel	the only member spanning the full width
Sash	20 × 20 × 1.6 aluminium	carried at the jambs by the stays
Screen	10 × 10 aluminium	in every rail of two or more segments

Transom check for this opening: it deflects 6.2 mm at 0.7 kPa, limit span/175 = 9.5 mm.

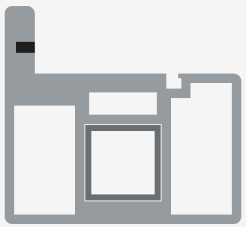
Segment joints are built to move



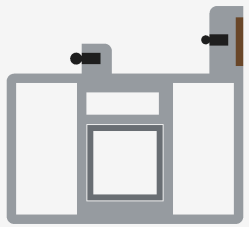
Schematic section through one joint at the cord ring (not to scale).

- PETG and ASA expand by 65–95 µm/m·K. A 240 mm segment grows **0.8–1.1 mm** over a 50 °C swing; one long printed part would bow instead.
- Each joint is assembled at about 20 °C on two **1 mm shims**. The generator shortens every segment by one gap per joint, so member lengths and tube cuts don't change.
- One ring of **4 mm closed-cell foam cord** sits in the 1.5 mm grooves of the two end faces.
- **MS polymer rated for ±50 %** fills the band outside the cord and the end faces of the stops and fins: the gap cycles between about 0 and 1.3 mm.

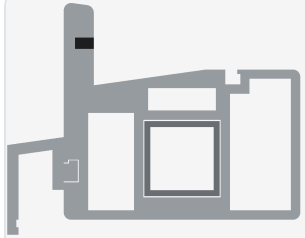
Profiles, thermal break and air chambers



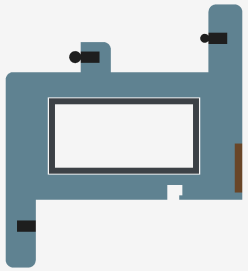
Glazing
lower jambs `JL`, sill-corner legs



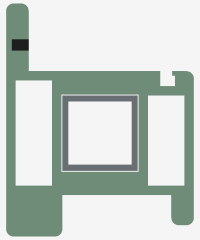
Sash rebate
head `H`, upper jambs `JU`, `CH`



Sill + nose
`S` `SW` and the clip-on `NS`



Transom
`T` `TE` and shoes `TSL` `TSR`



Sash
`SH` `SB` `SBE` `SV` `CS`



Screen
`NH` `NV` `CN`



Bead, IGU / acrylic
`BFH` `BFV` `BSH` `BSV`

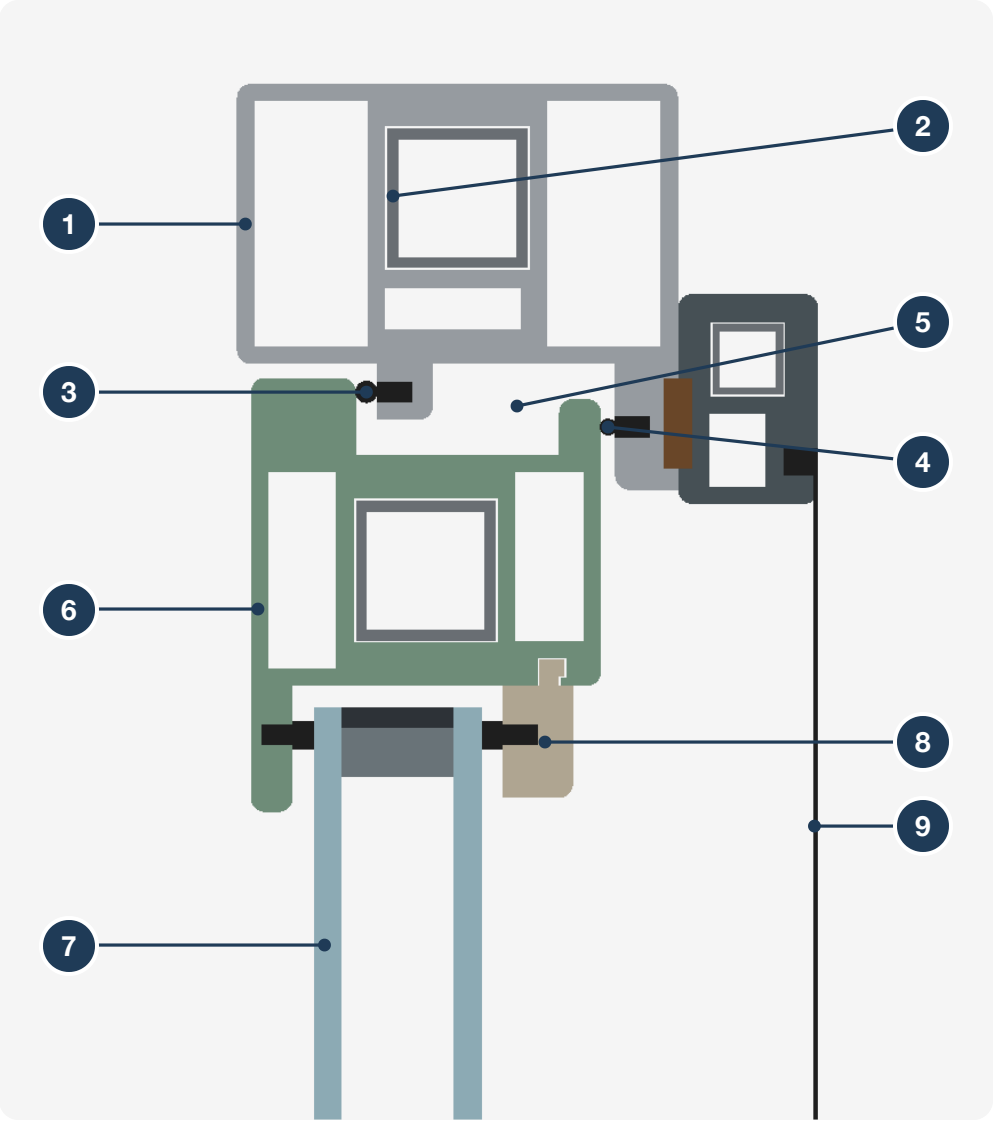


Bead, twin-wall
packer variant

All drawn at the same scale, outside on the left. Black: seals and gaskets; brown: magnetic strip.

- Every cross-section comes from one file, `cad/profile.scad`.
- The tube sits in a pocket **fully enclosed by plastic**, at least **6 mm** from the inside and outside faces.
- **No metal bridges** from outside to inside: stay and lock screws go into the core from one side only.
- Bodies are hollow, multi-chamber profiles with **2.5 mm walls**, like uPVC.
- Frame: 63 mm deep, 58 mm face. Transom body 34 mm, sash body 33 mm.

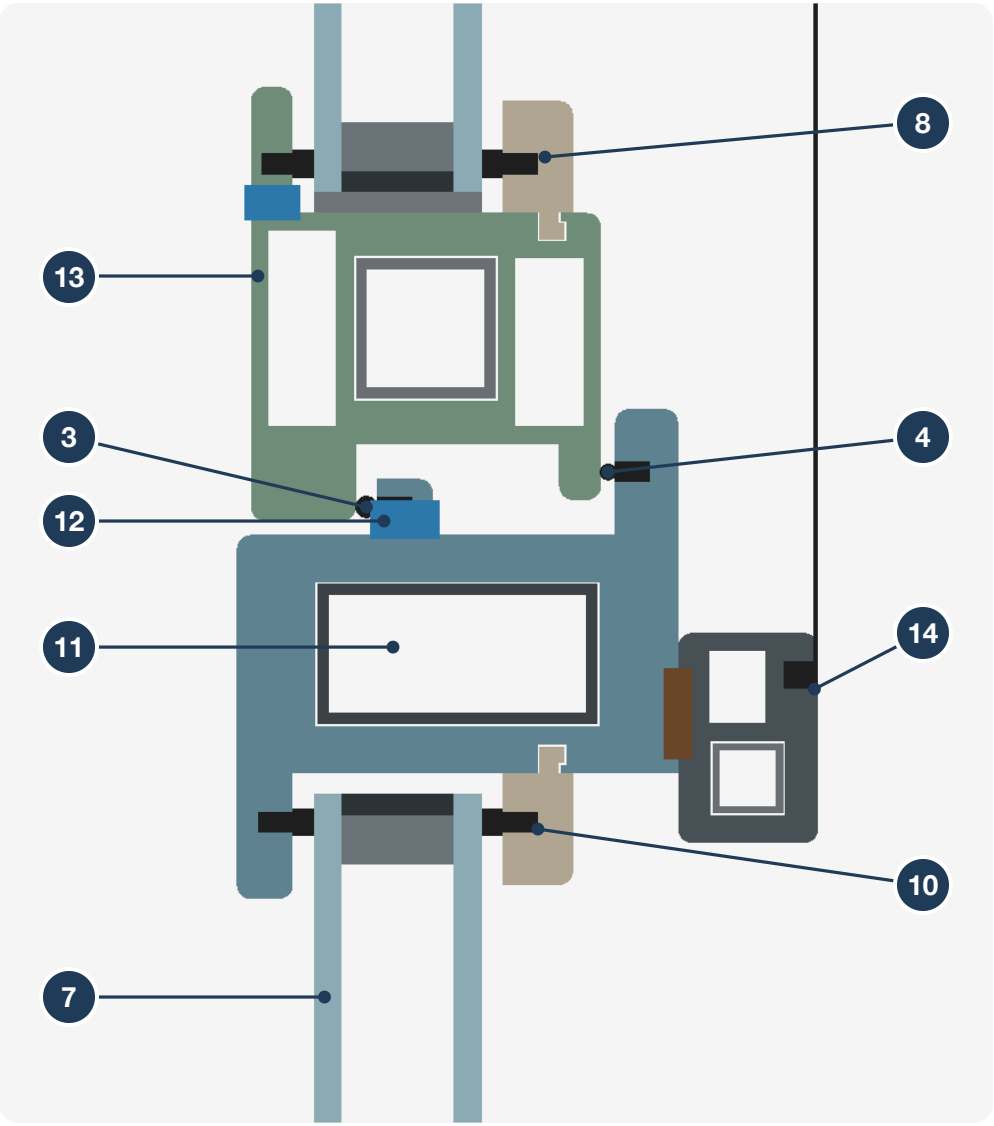
Section 1: head and top of the sash



Vertical section at mid-width, outside on the left.

1. Head: sash-rebate profile (`H` segments, `CH` corners)
2. 20 × 20 aluminium tube, wrapped in plastic
3. Outer weather seal: EPDM bulb in the fin kerf, closed on by the sash flange
4. Inner air seal: EPDM bulb in the inner-stop kerf, closed on by the sash lip
5. Pressure-equalised chamber between the seals: the 13 mm sash gap, where the stays live on the jambs
6. Sash top rail `SH` on its own 20 × 20 tube
7. 24 mm IGU between two 3 mm wedge gaskets
8. Snap-in bead `BSH` holds the inner gasket
9. Screen top rail `NH`: magnetic strip A against strip B in the frame channel; mesh in the spline groove

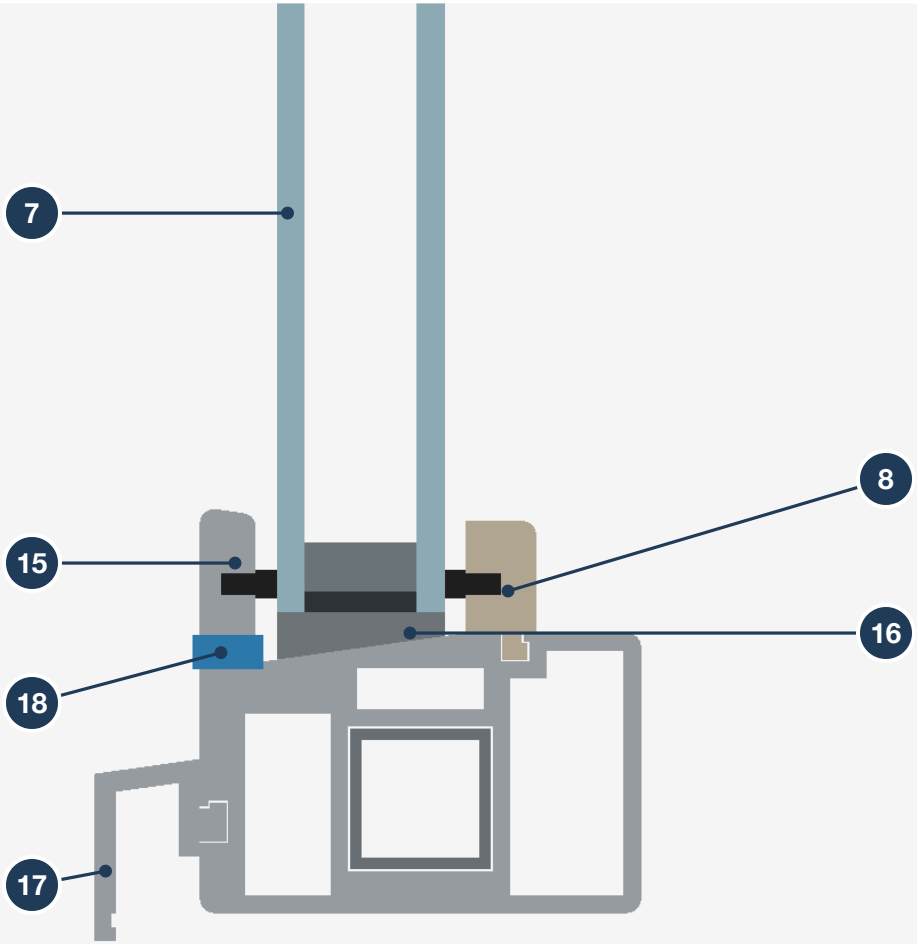
Section 2: transom, sash bottom and fixed-pane top



Blue: weep slots, cut only in `TE` and `SBE` segments.

10. Fixed-pane bead `BFH`, snapped into the transom's keyway
11. Transom `T` / `TE` on a 40 × 20 × 1.6 steel tube
12. Weep slot through the fin root, in the two `TE` segments: drains the chamber between the seals
3. Outer weather seal on the transom fin
4. Inner air seal on the transom's inner stop
13. Sash bottom rail `SB` / `SBE`: glass on `GF` setting blocks; the `SBE` weeps drain the sash pocket
14. Screen bottom rail on the lower half of the transom's inside face; the cam locks sit above it, inside the closed screen
7. Fixed IGU (below) and sash IGU (above)
8. Sash bead `BSH`

Section 3: sill and bottom of the fixed pane



15 Sill `S` / `SW`: the glazing-pocket floor falls 8° outward to the weep

16 `GW` wedge setting block at each quarter point, over a packer under the sill

17 Clip-on sill nose `NS` with drip groove, 15 mm proud of the frame face

18 Weep slot through the outer stop (`SW` segments), behind a flap cap `WC`

7 Fixed IGU standing on its setting block

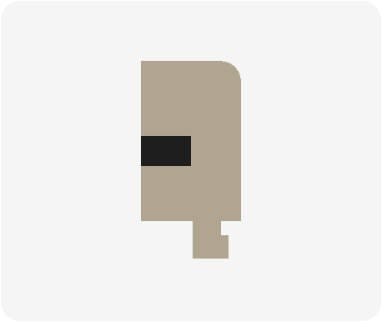
8 Bead `BFH`, snapped in from the room side

> ! Never seal over the weep slots or the nose drip. Water that gets past a seal must have a way out.

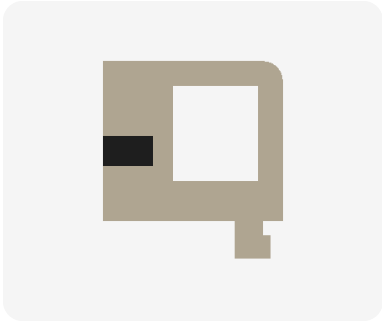
Glazing: one 24 mm pocket, three fillings

Glazing	Make-up	Bead	Notes
IGU (glazier)	4/16/4 or 5/14/5, low-E/argon optional	standard	best thermal and clarity; heaviest
Acrylic sandwich	4 mm + 16 mm printed spacer + 4 mm	standard	spacer has a desiccant channel; unit can be opened to clean
Twin-wall PC	16 mm sheet + 8 mm printed packer	packer	translucent: bathrooms, laundries

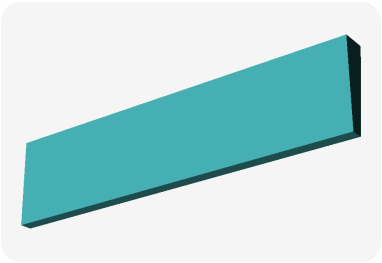
- 3 mm EPDM **wedge gaskets** on both faces: the outer one in the stop kerf, the inner one in the bead.
- Glass stands on printed **setting blocks**: `GW` wedges on the sloped sill, `GF` flat blocks in the sash and as side locators.
- **Beads snap** into an undercut keyway from the room side; no screws.
- Each glazing pocket is drained and ventilated through weep holes with flap caps.



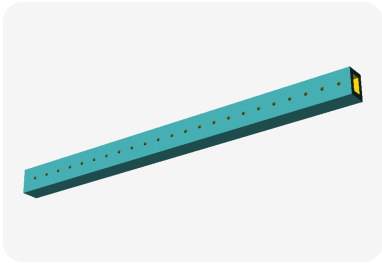
IGU / acrylic bead



Twin-wall bead

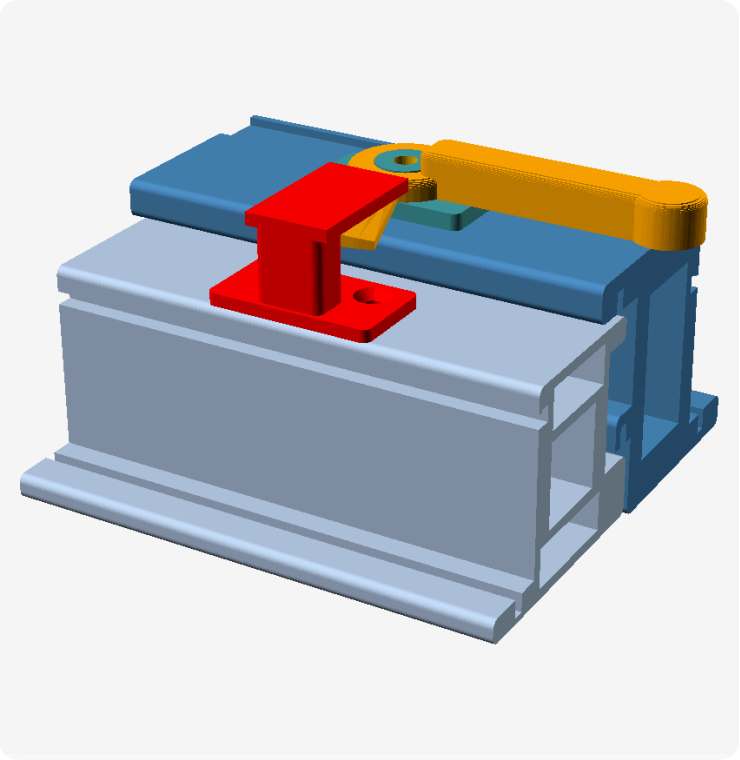


`GW` wedge block

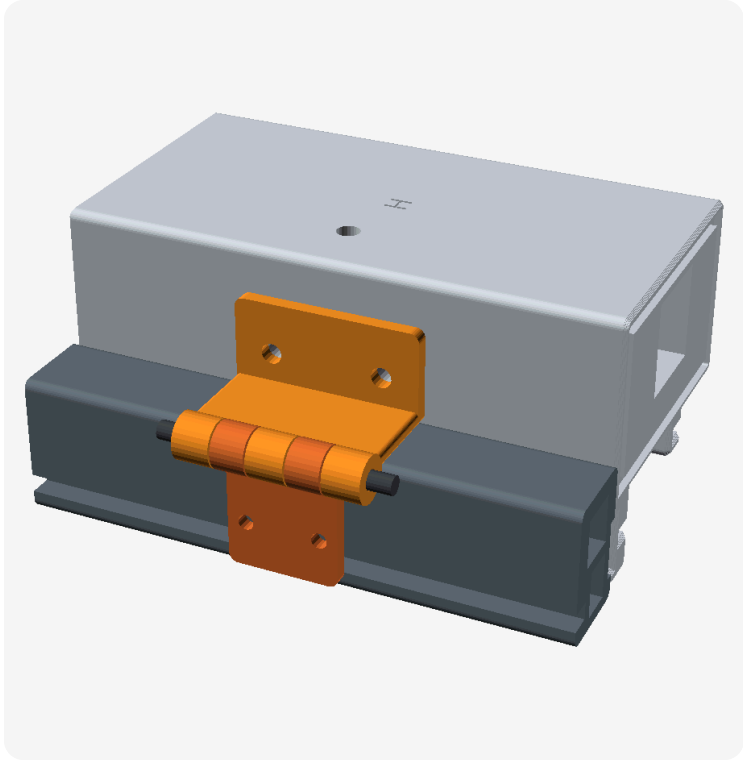


`AFH` acrylic spacer

Hardware: cam locks, screen hinges, friction stays



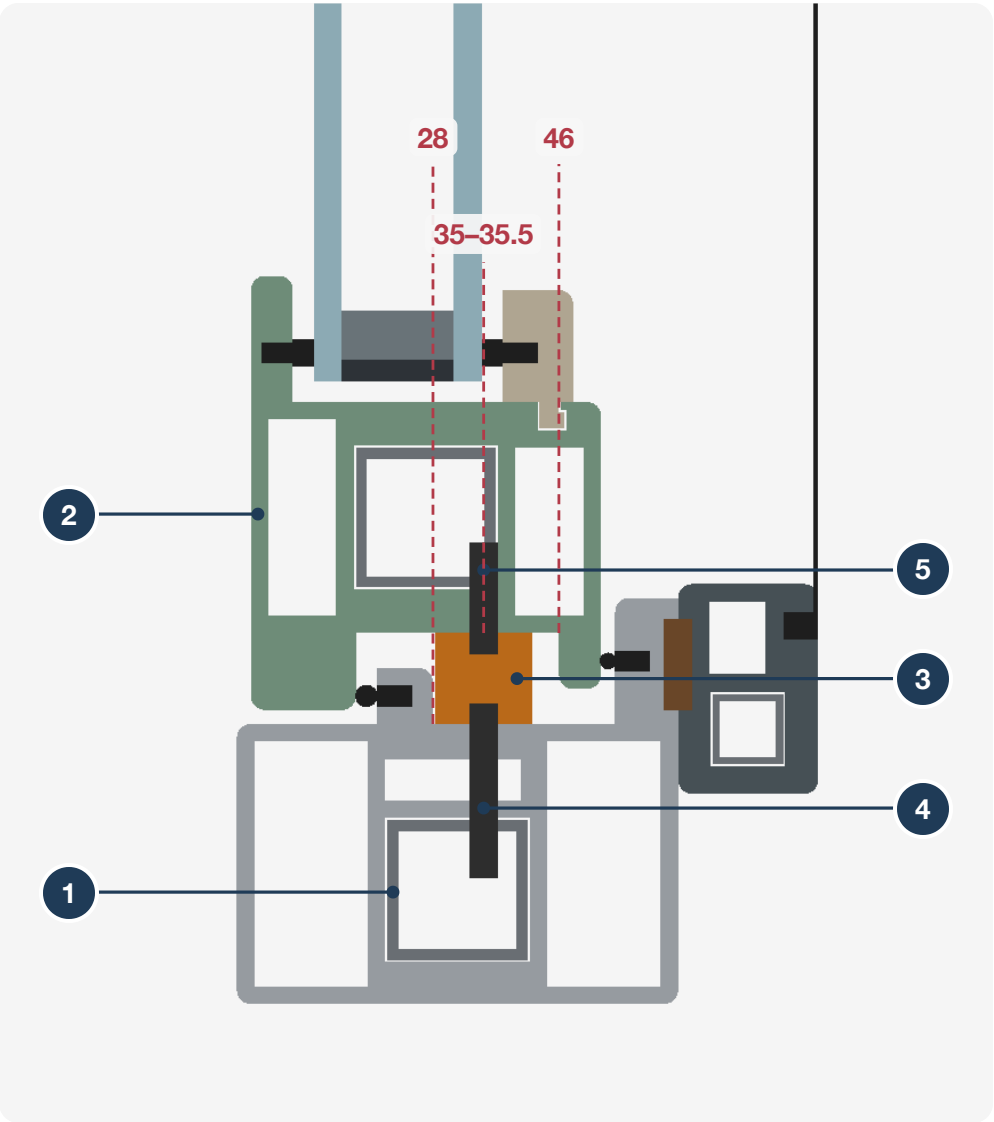
Cam lock: base `CMB` and handle `CMH` on a `TE`, keeper `KP` on the sash



Screen hinge: `HG` on an `H` segment, `HS` on the screen rail, M4 × 50 pin

- **Two cam locks**, one near each end: turning the lever 60° pulls the sash in about 3 mm onto both seals. Each keeper sits over its cam pivot.
- **Screen:** 3 pin hinges on the head; it closes under gravity onto magnetic strips A and B.
- **Friction stays:** a stainless pair holds the sash at any opening. Screws go into the tubes from one side only.
- Heat-set M4 inserts take every machine screw that gets re-tightened.

The friction-stay gate: check this before printing



1. Upper jamb `JU` and its tube (21.5–41.5 mm from the outside face; V-slot cores: see the V-slot slide)
2. Sash side `SV` and its tube (17–37 mm from the sash's outside face)
3. Folded stay: **≤ 13 mm stack** in the sash gap, **≤ 14 mm wide**, centred 35–35.5 mm from the outside face
4. Track screws 8g × 25 into the jamb tube
5. Arm screws 8g × 16 into the sash tube

Dimensions in mm from the frame's outside face: the stay must sit between the jamb fin top (28) and the sash lip (46).

! Find a 400 mm stay pair rated ≥ **15.7 kg** that fits this envelope before printing anything full size. Most 400 mm awning stays are 16–19 mm wide and will not fit.

Pick the material tier for your wall

Tier	Exposure	Min HDT	UV	Examples
1	South-facing or shaded, light colour	≥ 70 °C	stable or painted	PETG + exterior acrylic paint, ASA
2	East or part sun, light colour	≥ 90 °C	inherently stable	ASA
3	North or west full sun, or any dark colour	≥ 105 °C	inherently stable	ASA-CF, PC-ABS (painted)

- **Rule:** heat-deflection temperature (HDT at 0.45 MPa) at least **15 °C above** the hottest surface, and exterior parts UV-stable or painted.
- Melbourne design surface temperatures: about 50–55 °C on light colours, 75–80 °C on dark colours in afternoon sun.
- **A dark colour moves any exposure up one tier.** Light colours are strongly recommended.
- Set `tier` in your params file; the BOM names the material and weighs it.

Exterior parts (tier material)

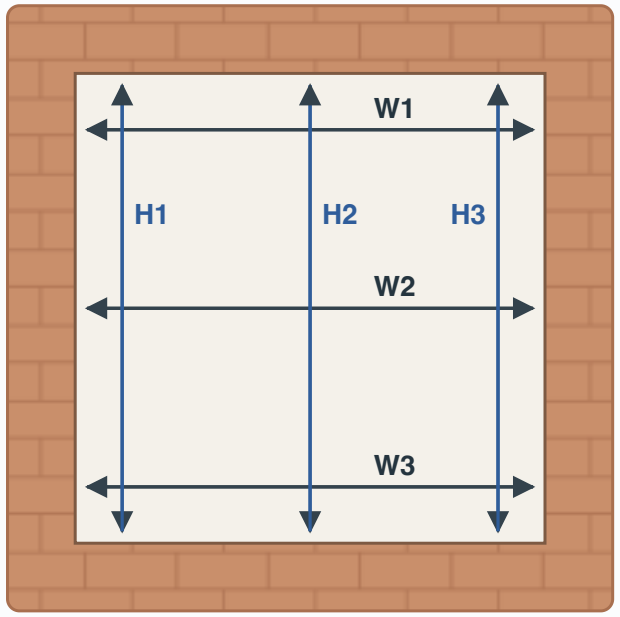
`H` `S` `SW` `JL` `JU` `CH` `CSL` `CSR` `JN` `NS` `T` `TE` `TSL` `TSR` `SH` `SB` `SBE` `SV` `CS` `TC` `WC` `GW` `GF`

Interior parts (PETG in every tier)

`NH` `NV` `CN` `HG` `HS` `CMB` `CMH` `KP` `BFH` `BFV` `BSH` `BSV`

ⓘ P1S: ASA and PC need the enclosure closed; CF filaments need a hardened-steel nozzle.

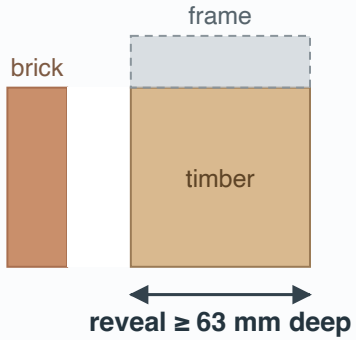
Measure the opening



Width, reveal to reveal:
W1, W2, W3

Height, head to brick sill:
H1, H2, H3

Use the smallest of each
for opening_w and opening_h



Frame = opening – 2 × fix_gap. The home window's 1800 × 1500 opening gives a 1784 × 1484 frame.

1. Take the old window out first (install step 1), or measure inside its frame and confirm later.
2. Measure **width** reveal to reveal and **height** head to brick sill, three times each. Use the smallest.
3. Check the timber reveal is at least **63 mm deep**, so the frame sits fully on it.
4. Pick `fix_gap`, the gap to the brickwork each side: **6–10 mm**.

! Print and order glass only from these measurements, after regenerating the BOM.

Your params file: six numbers and a name

Field	Meaning	Home	Bench
`name`	must match the file name	`home-1800x1500`	`bench-600`
`opening_w`	measured width (mm)	`1800`	`600`
`opening_h`	measured height (mm)	`1500`	`600`
`fix_gap`	gap to the brickwork each side, 6–10	`8`	`8`
`awning_h`	awning height, at least 150	`450`	`200`
`glazing`	`igu`, `acrylic` or `twinwall`	`igu`	`acrylic`
`tier`	material tier 1–3	`3`	`1`

```
cp params/home-1800x1500.json params/my-window.json
```

Then edit it:

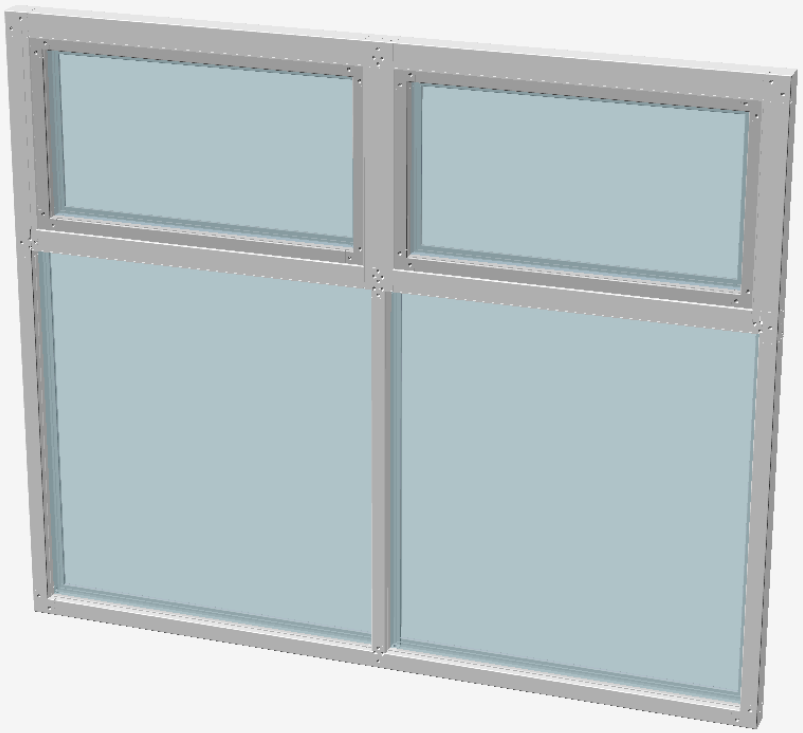
```
{
  "name": "my-window",
  "opening_w": 1800,
  "opening_h": 1500,
  "fix_gap": 8,
  "awning_h": 450,
  "glazing": "igu",
  "tier": 3
}
```

Sheet smaller than the panes? Mullions and bays

Add `"max_pane": [1200, 900]` to the params file: the largest glass or acrylic sheet you can get, either way round.

- The generator picks the **fewest equal bays** that make every pane fit, splits the window with **mullions**, and gives each bay its own fixed pane, awning sash, screen and two cams.
- `"bays": n` asks for more bays than needed, never fewer.
- A pane too tall for the sheet needs a second transom, which OpenWindow does not do: the generator stops and says so.
- New parts: sill node `SN`, head node `HN`, transom cross-node `TX`, mullion shoes `MSL`, `MST`, `MSH` and segments `ML`, `MU`, each on a 40 × 20 steel tube.

	One bay	1200 × 900 mm sheet
Fixed panes	1 × 1698 × 878	2 × 829 × 878
Sashes	1, 15.7 kg	2, 7.6 kg each
Transom	6.2 mm at 0.7 kPa, limit span/175 = 9.5 mm	6.8 mm at 0.7 kPa carrying 1 mullion, limit span/175 = 9.5 mm
Mullion	none	lower mullion deflects 0.6 mm at 0.7 kPa over 848 mm, limit span/175 = 4.8 mm
Prints	160 in 35 types	226 in 43 types
Filament	16.36 + 3.84 kg	19.06 + 4.75 kg



`params/home-1800x1500-2bay.json`, seen from the room.

Generate, read the checks, render, build the BOM


```
$ python3 tools/generate.py params/home-1800x1500.json
home-1800x1500: 35 part types, 160 prints -> cad/generated
ok       transom deflects 6.2 mm at 0.7 kPa, limit span/175 = 9.5 mm
ok       sash 15.7 kg, stay pair 400 mm rated 30.0 kg
ok       segments over bed: []
ok       keeper bosses 39.43 mm either side of the SBE middle, over the cams (allowed 20.6–91.7 mm, transom needs 2+ segments)
```

- **Transom deflection** against span/175 at 0.7 kPa wind.
- **Sash weight** against the friction-stay pair it picks; the BOM asks for a stay rated at least this weight.
- **Segment length** ≤ 240 mm, and **cam/keeper** alignment.

! A `WARNING` line needs a change before you print.

```
cad/box.sh render my-window
python3 bom/build_bom.py my-window
```

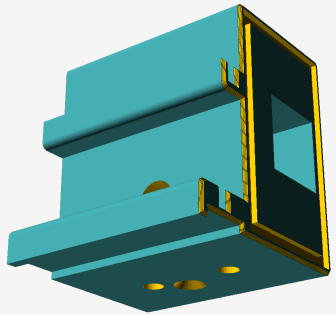
- `box.sh` renders every part on an OpenSCAD machine and checks each STL against the part list. No such machine? Export each `cad/generated/parts/<ID>.scad` in your own OpenSCAD.
- The BOM step writes `bom/my-window.md` and the `-parts`, `-cutlist` and `-glass` CSVs.

i Build from the per-name files in `bom/`: `cad/generated/` only holds the last window generated.

Printer settings (Bambu P1S)

- 0.4 mm nozzle (hardened steel for CF filaments), **0.2 mm layers**.
- **5 walls, 5 top and 5 bottom layers, 15 % gyroid infill.** Most profile walls are 2.5 mm, so they print solid anyway.
- ASA and PC-ABS: enclosure closed, door shut, no auxiliary fan, textured PEI plate with glue stick.
- Exterior parts in your tier's material, interior parts in PETG (the parts list marks each).

✓ Print the test coupon `TC` and one segment of each profile first, and check the fit before committing a full spool.



Test coupon `TC`: a 60 mm sash-rebate segment with two inserts.

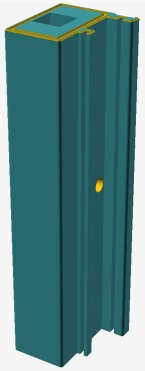
Heat-set inserts: 30 × M4 for this window

Pressed flush with a 230 °C iron, wherever a machine screw is re-tightened:

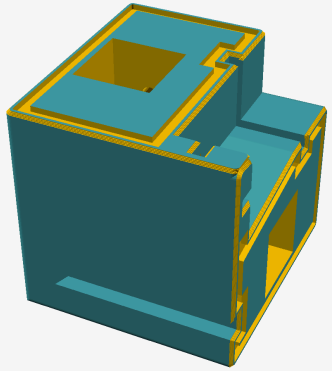
- `H`: 2 each, on every head segment, so the hinges can move later;
- `TE`: cam bases;
- `SBE`: both boss pairs, 4 each (the keeper uses the outer pair);
- `CMB`: the cam pivot boss;
- `TC`: 2, for the heat and torque test.

Orientation on the bed

Parts	Orientation
Segments `H` `S` `SW` `JL` `JU` `T` `TE` `SH` `SB` `SBE` `SV` `NH` `NV`, and `TC`	Standing on an end face, exactly as exported. No supports. The 2 mm rounded long edges stand vertical, and each 0.8 mm joint bevel prints as four 0.2 mm layers.
Corners `CH` `CSL` `CSR` `CS` `CN`	As exported. Supports on the build plate only, for the rebate overhangs.
`JN` `TSL` `TSR`	On the outside face, so the dovetail slot and tongue print in plane and slide without binding. Supports on the plate only.
Beads (B…)	On their glass face (the gasket-kerf side), so the rounded edge prints on top.
Sill nose `NS`, acrylic spacers (A…)	Laid flat on their longest face.
Setting blocks, weep caps, cam parts, hinge leaves, keepers	Flat side down.



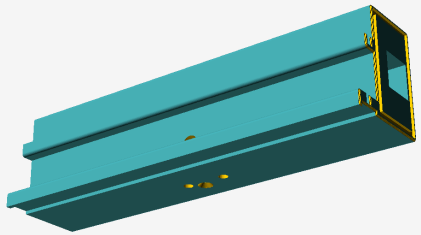
`H` as exported: standing on its end face.



`CSL` as exported.

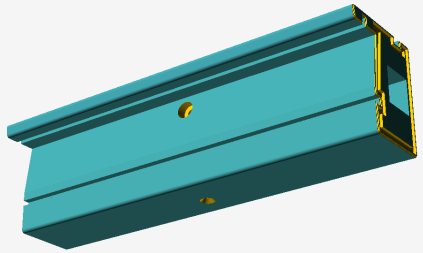
> ℹ The catalogue that follows shows every part lying down, so its profile is visible; print segments and corners as exported.

Part catalogue (1/4)



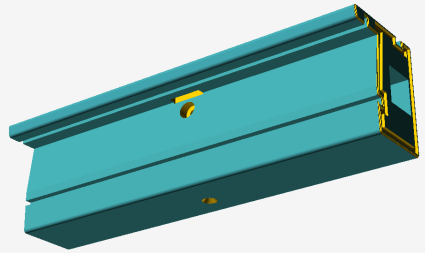
H Head segment, hinge inserts

● 7 × home 305 g



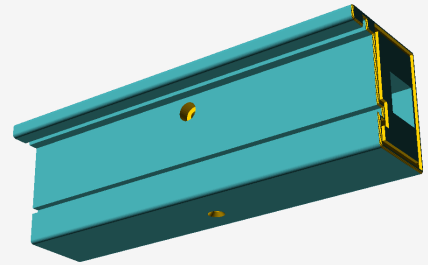
S Sill segment

● 5 × home 337 g



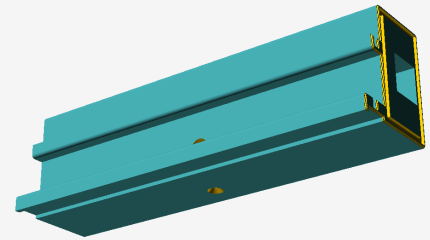
SW Sill segment with weep

● 2 × home 336 g



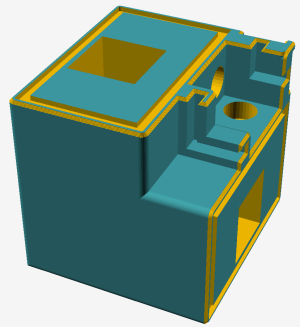
JL Lower jamb segment

● 8 × home 287 g



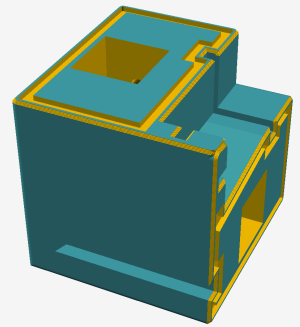
JU Upper jamb segment

● 4 × home 287 g



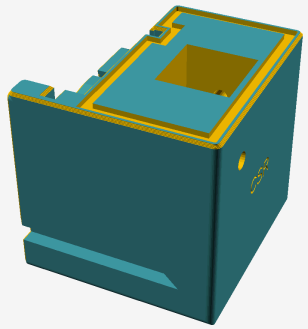
CH Head corner

● 2 × home 115 g



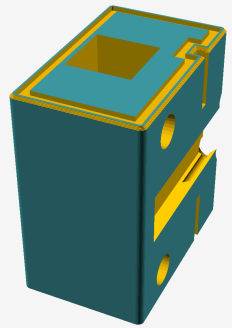
CSL Sill corner, left

● 1 × home 120 g



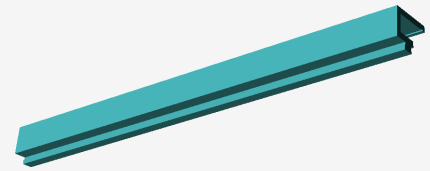
CSR Sill corner, right

● 1 × home 120 g



JN Jamb node (transom slot)

● 2 × home 85 g

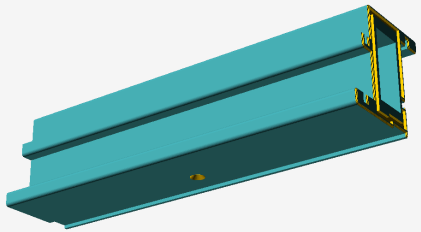


NS Sill nose

● 8 × home 42 g

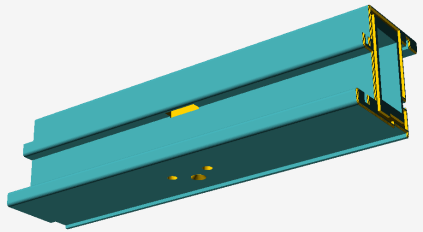
● exterior, tier material ● interior, PETG · quantity for the home window (acrylic spacers: bench window) · mass each, estimated from the STL. Not to scale.

Part catalogue (2/4)



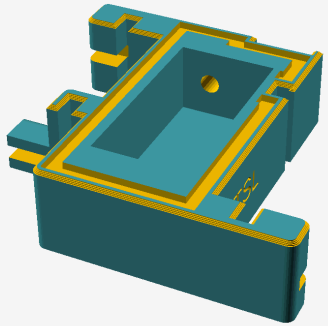
T Transom segment

● 5 × home 301 g



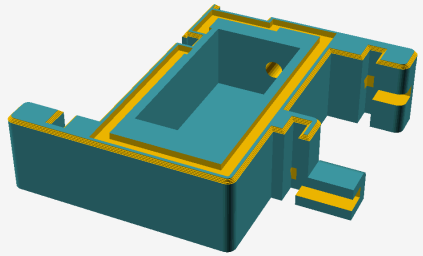
TE Transom end: weep, cam inserts

● 2 × home 302 g



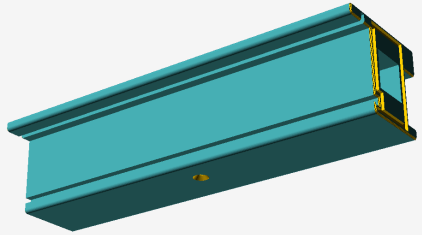
TSL Transom shoe, left

● 1 × home 38 g



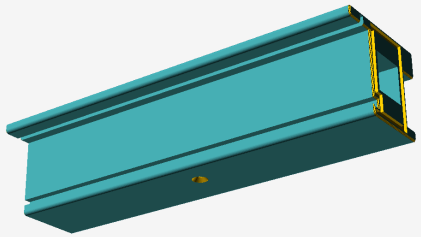
TSR Transom shoe, right

● 1 × home 38 g



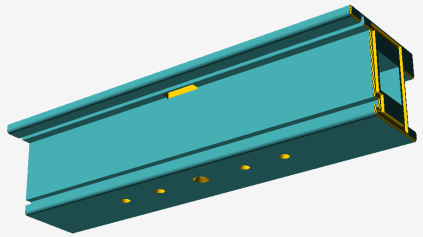
SH Sash top segment

● 7 × home 274 g



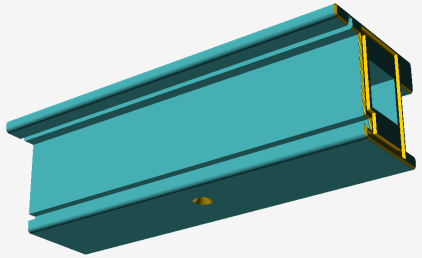
SB Sash bottom segment

● 5 × home 274 g



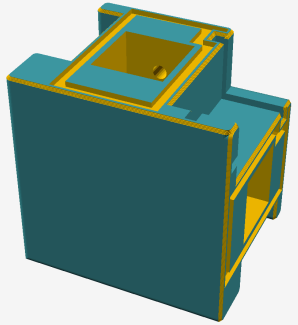
SBE Sash bottom end: weep, keeper inserts

● 2 × home 276 g



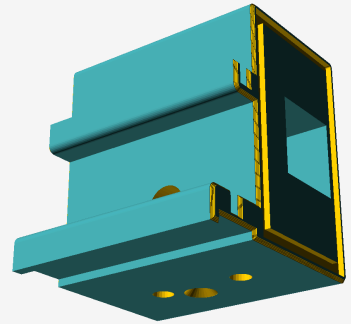
SV Sash side segment

● 4 × home 218 g



CS Sash corner

● 4 × home 92 g

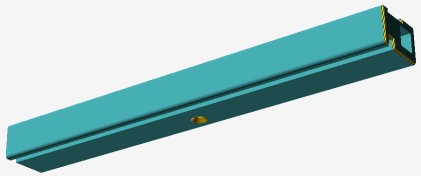


TC Test coupon

● 1 × home 90 g

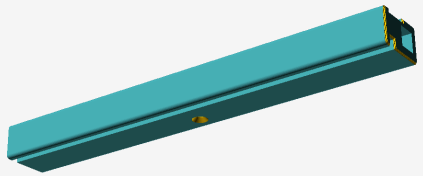
● exterior, tier material ● interior, PETG · quantity for the home window (acrylic spacers: bench window) · mass each, estimated from the STL. Not to scale.

Part catalogue (3/4)



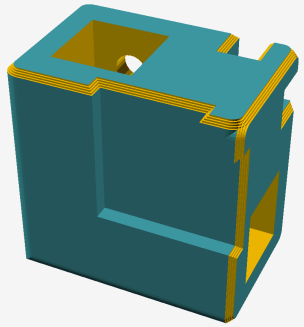
NH Screen horizontal segment

● 14 × home 108 g



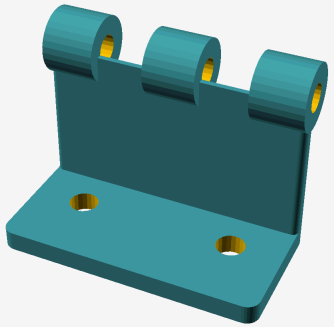
NV Screen vertical segment

● 4 × home 109 g



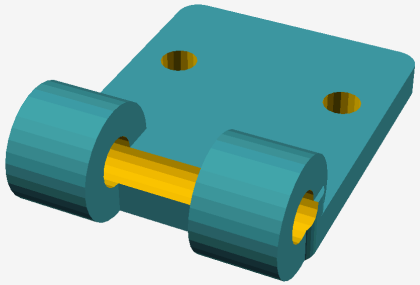
CN Screen corner

● 4 × home 16 g



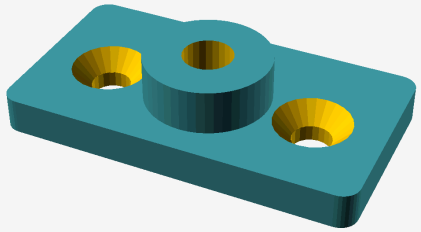
HG Hinge leaf, frame

● 3 × home 8 g



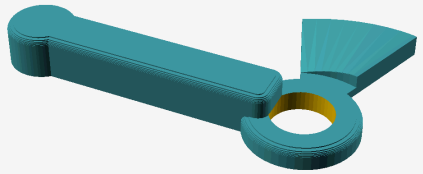
HS Hinge leaf, screen

● 3 × home 4 g



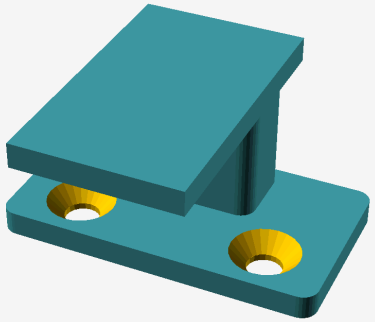
CMB Cam base

● 2 × home 5 g



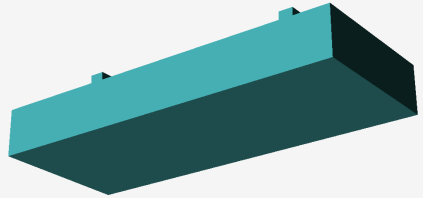
CMH Cam handle

● 2 × home 14 g



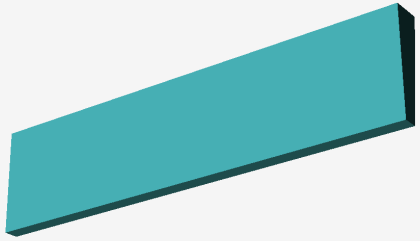
KP Cam keeper

● 2 × home 9 g



WC Weep cap

● 4 × home 1 g

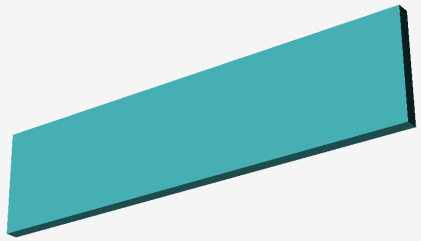


GW Wedge setting block

● 2 × home 15 g

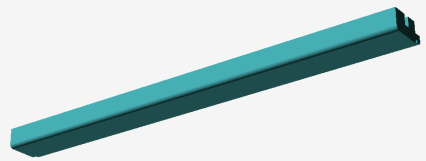
● exterior, tier material ● interior, PETG · quantity for the home window (acrylic spacers: bench window) · mass each, estimated from the STL. Not to scale.

Part catalogue (4/4)



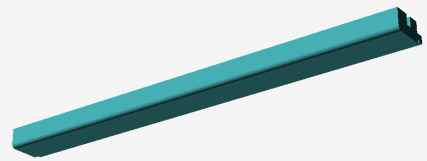
GF Flat setting block / locator

● 6 × home 9 g



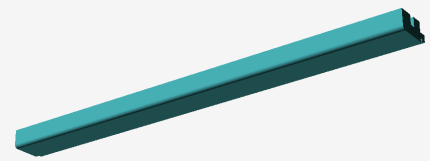
BFH Fixed-pane bead, horizontal

● 16 × home 41 g



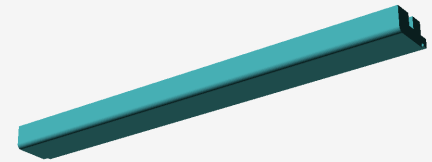
BFV Fixed-pane bead, vertical

● 8 × home 41 g



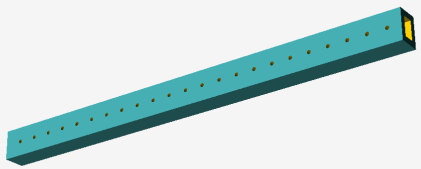
BSH Sash bead, horizontal

● 14 × home 44 g



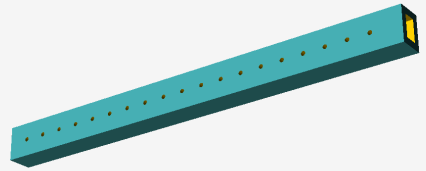
BSV Sash bead, vertical

● 4 × home 35 g



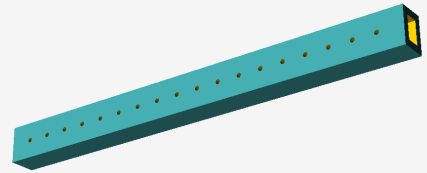
AFH Acrylic spacer, fixed, horiz.

● 4 × bench, acrylic only 39 g



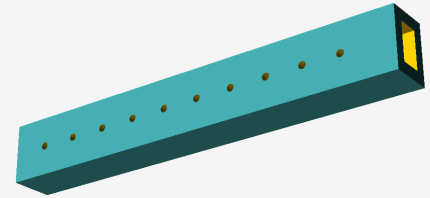
AFV Acrylic spacer, fixed, vert.

● 2 × bench, acrylic only 34 g



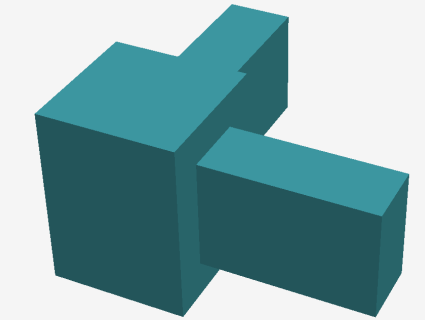
ASH Acrylic spacer, sash, horiz.

● 4 × bench, acrylic only 32 g



ASV Acrylic spacer, sash, vert.

● 2 × bench, acrylic only 19 g



AC Acrylic spacer corner

● 8 × bench, acrylic only 5 g

● exterior, tier material ● interior, PETG · quantity for the home window (acrylic spacers: bench window) · mass each, estimated from the STL. Not to scale.

Filament and print time (home window)

Family	Parts	Prints	kg
Frame segments	H S SW JL JU	26	7.93
Frame corners, nodes, nose	CH CSL CSR JN NS	14	0.97
Transom	T TE TSL TSR	9	2.19
Sash	SH SB SBE SV CS	22	5.09
Beads	BFH BFV BSH BSV	42	1.75
Screen and hinges	NH NV CN HG HS	28	2.04
Small parts	GW GF WC CMB CMH KP TC	19	0.23
Total		**160**	**20.20**

16.36 kg
exterior, tier material

3.84 kg
interior PETG

≈ 550 h
printer time (spec §7 estimate, from the STLs)

Masses are estimated from the rendered STLs with the slicer settings above (2.1 mm shell, 15 % infill), as in `bom/build_bom.py`.

Bill of materials, home window (1/3): fixings

Group	Item	Qty	Unit
Fixings	Timber screw 10g x 75, stainless or galvanised (frame fixings)	38	pcs
Fixings	Self-tapping screw 8g x 30 stainless, countersunk (aluminium tube screws: frame, sash, corners, nodes)	62	pcs
Fixings	Tek screw 8g x 25, 410 stainless, countersunk (steel transom tube: T, TE, shoes)	9	pcs
Fixings	Self-tapping screw 8g x 12 stainless, countersunk (screen tube screws)	26	pcs
Fixings	M4 heat-set insert for a 5.6 x 8 mm hole	30	pcs
Fixings	M4 x 10 countersunk machine screw (cam bases CMB, keepers KP)	8	pcs
Fixings	M4 x 10 pan or button head machine screw (hinge frame leaves HG)	6	pcs
Fixings	M4 x 8 pan screw + 20 mm penny washer (cam pivots)	2	pcs
Fixings	M4 x 16 countersunk machine screw (transom lock screws)	4	pcs
Fixings	M4 x 50 bolt + nyloc nut (hinge pins)	3	pcs
Fixings	4g x 12 self-tapping screw (screen leaves HS)	6	pcs
Fixings	Plastic window packers, assorted 1-10 mm (one stack per position)	44	pcs

Every line is generated from the geometry by `bom/build_bom.py`. Your window: `bom/<name>.md`.

Bill of materials, home window (2/3): hardware and seals

Group	Item	Qty	Unit
Hardware	Stainless friction stay pair, 400 mm, stack <= 13 mm, track/arm width <= 14 mm (check rating >= 15.7 kg)	1	pair
Hardware	Friction-stay screw 8g x 25 stainless, countersunk (tracks to upper jambs; match the stay's holes)	8	pcs
Hardware	Friction-stay screw 8g x 16 stainless, countersunk (arms to sash sides; match the stay's holes)	6	pcs
Seals	EPDM bulb kerf seal, 3 mm kerf (outer + inner loop round the sash opening)	8.92	m
Seals	EPDM glazing wedge gasket, 3 mm	18.28	m
Seals	Closed-cell foam cord, 4 mm (one ring per segment joint)	11.3	m
Seals	Closed-cell foam cord, 3 mm (transom shoe seams)	0.14	m
Seals	MS-polymer sealant rated for +/-50% joint movement (ASTM C920 Class 50 or equal), 290 ml cartridge (joints, fixing bedding, outside perimeter)	3	pcs
Seals	Butyl flashing tape, 150 mm wide (sill tray; or two 100 mm runs lapped 50 mm)	2.08	m
Seals	Backer rod, 10 mm	6.54	m
Seals	Low-expansion window foam, can	1	pcs
Seals	Gel CA glue (EPDM loop joints)	1	pcs

Bill of materials, home window (3/3): screen, tube, glass, filament

Group	Item	Qty	Unit
Screen	Flexible magnetic strip A, 12.7 mm, adhesive-backed (screen back channel)	4.53	m
Screen	Flexible magnetic strip B, 12.7 mm, adhesive-backed, matched to A, or adhesive steel strip (frame magnet channel)	4.53	m
Screen	Fibreglass insect mesh	1.02	m2
Screen	Screen spline, 4 mm	4.53	m
Screen	Two-part epoxy, 5-minute (screen corner joints)	1	pcs
Tube	Tube alu10 (total length; buy in 1 m/3 m stock)	4.37	m
Tube	Tube alu20 (total length; buy in 1 m/3 m stock)	10.31	m
Tube	Tube steel40x20 (total length; buy in 1 m/3 m stock)	1.7	m
Glazing	IGU 4/16/4 = 24 mm, toughened if required by AS 1288, low-E/argon optional, 1698 x 878 (fixed pane)	1	pcs
Glazing	IGU 4/16/4 = 24 mm, toughened if required by AS 1288, low-E/argon optional, 1606 x 388 (sash pane)	1	pcs
Filament	PETG (interior)	3.84	kg
Filament	ASA-CF or PC-ABS (light colour) (exterior, tier 3)	16.36	kg

Tube cut list (home window)

Member	Core	Length (mm)	Qty
head	alu20	1721	1
sill	alu20	1721	1
jamb	alu20	1421	2
transom	steel40x20	1696	1
sash horizontal	alu20	1615	2
sash vertical	alu20	397	2
screen horizontal	alu10	1685	2
screen vertical	alu10	501	2

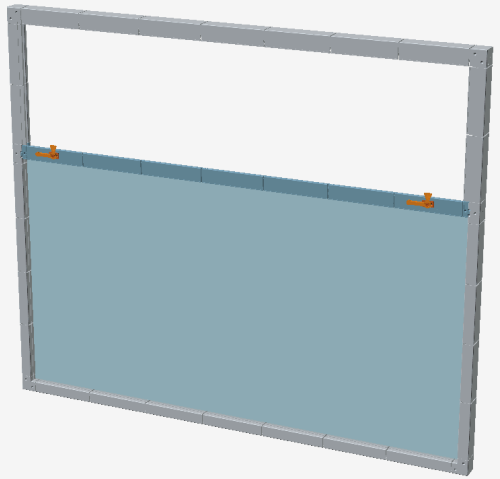
- Cut with a hacksaw or a drop saw with a metal blade, **never longer than the list**. Each length leaves 1.5 mm clear at every corner-socket floor for print shrinkage.
- Dry-fit each member on its tube with the shims in. Trim the tube if a joint won't close onto its shims.
- Deburr and wipe with isopropyl alcohol.
- Buy aluminium in 1 m or 3 m stock: 10.31 m of 20 × 20 × 1.6 and 4.37 m of 10 × 10. Steel: 1.7 m of 40 × 20 × 1.6 galvanised.
- The transom tube is checked again at install step 6 against the real node spacing.

Glass order (home window)

Pane	Width (mm)	Height (mm)	Qty
Fixed	1698	878	1
Sash	1606	388	1

Spec: IGU 4/16/4 = 24 mm, toughened if required by AS 1288, low-E/argon optional.

Weights: fixed about 30 kg, sash glass about 12 kg.



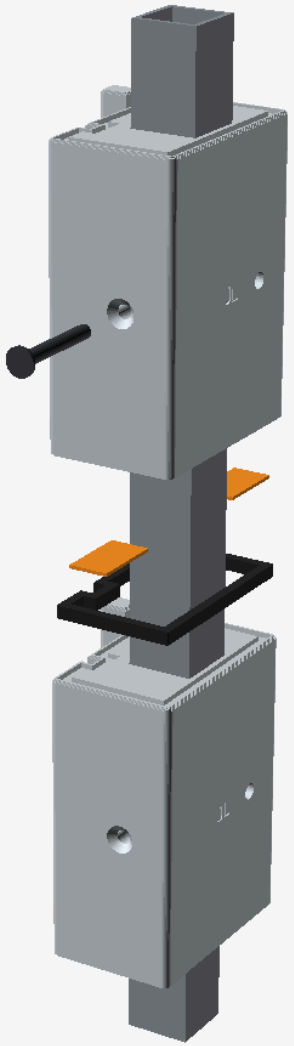
The fixed IGU sits below the transom; the sash IGU is glazed at the bench.

- Order from `bom/<name>-glass.csv` only **after** the opening is measured and the BOM rebuilt.
- Your glazier decides where AS 1288 requires toughened or laminated glass.
- Acrylic glazing: two 4 mm sheets per pane, cut to the same sizes, film on until assembly (bench window).
- Twin-wall: 16 mm polycarbonate, UV face out, flutes vertical.

Tools you will need

- **Cutting:** hacksaw or drop saw with a metal blade, file or deburring tool, utility knife
- **Drilling:** drill/driver; bits 2 mm (screen leaves), 3.2 mm (aluminium tube and stay pilots), 3.5–3.8 mm (steel transom, if not using tek screws); countersink to tidy holes
- **Driving:** Phillips/square driver bits for 4g, 8g and 10g screws, hex keys for M4, a small spanner for the nylocs
- **Inserts:** soldering iron with a heat-set insert tip, set to 230 °C
- **Joints:** caulking gun, sealant spatula, bar or F clamps, 1 mm shims (plastic strips or feeler leaves) and 0.5–1.5 mm shims for the transom
- **Measuring:** tape, steel rule, square, spirit level (plumb and diagonals within 2 mm)
- **Seals and screen:** spline roller, scissors, gel CA, 5-minute epoxy
- **Install:** pry bar and knife for the old window, foam gun or can, backer-rod tool or blunt putty knife, ladder or platform, glass suction lifters
- **Safety:** gloves, eye protection, and a second person for lifting
- **Testing (bench):** garden hose, smoke pencil and fan, heat lamp and thermocouple, torque screwdriver (1.5 N·m)

Joint rules (1): temperature, gap, cord



Frame, transom and sash joints all follow these rules. Screen joints close tight with no shims and no cord.

- **Temperature:** assemble at about 20 °C (15–25 °C).
- **Gap:** every segment joint gets a 1 mm gap; the generator already shortened each segment by one gap per joint, so the member keeps its length.
- **Shims:** hold each gap with two 1 mm shims (strips of 1 mm plastic sheet, or feeler-gauge leaves), slipped in from the outside and inside faces. Pull them once that part's tube screw is in.
- **Cord:** one ring of 4 mm closed-cell foam cord per joint, pressed into the groove of **one** face only; the mating face's groove takes the other half.

Joint rules (2): sealant and tube screws

Sealant: MS polymer rated ±50 %

Butter it on the face before closing each joint, keeping it off the cord:

- over the whole **band outside the cord ring**, continuous with the end faces of the stops, fins and inner stop;
- around the **tube pocket**;
- on transom faces, across the 2.7 mm **web** between the bead keyway and the tube.

Work joint by joint, before it skins (about 10 min)

1. Butter, close onto the shims, clamp, drive that part's tube screw.
2. Pull the shims; fill and tool their voids, flush with the 0.8 mm bevel.
3. Clear squeeze-out from the kerfs, magnet channel and bead keyways.

Tube screws

One through the inside-face countersink of every segment, corner leg, node and shoe.

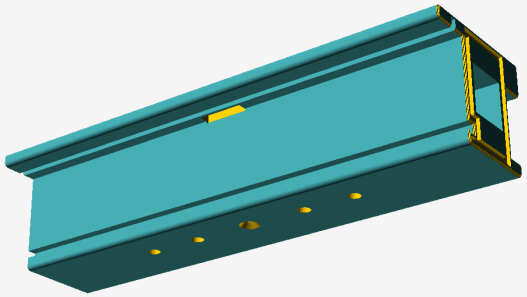
- **Aluminium tubes:** 8g × 30 stainless countersunk self-tappers, in a 3.2 mm pilot drilled through the printed hole into the tube.
- **Steel transom tube:** 8g × 25 tek screws in 410 stainless, or a 3.5–3.8 mm pilot and a stainless self-tapper.
- **V-slot cores:** M5 countersunk screws into slide-in T-nuts, with no pilots (see the V-slot slide).

! Stop when the head is flush, so the screw doesn't strip the plastic.

Step 1 · Prepare tubes and inserts · Step 2 · Acrylic sandwich

1 · Prepare

1. Cut the tubes to the cut list, never longer (V-slot cores: see the V-slot slide). Dry-fit each member with the shims in; trim if a joint won't close. Deburr, wipe with isopropyl.
2. Press M4 heat-set inserts (230 °C, flush) into every boss: all `H` segments (2 each), `TE`, `SBE` (both pairs, 4 each), the `CMB` pivot boss and `TC`.



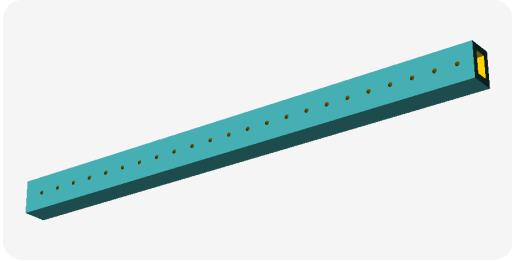
`SBE`: two insert pairs either side of the middle; the keeper uses the outer pair, nearer the sash corner.

2 · Acrylic sandwich (only if `glazing = acrylic`)

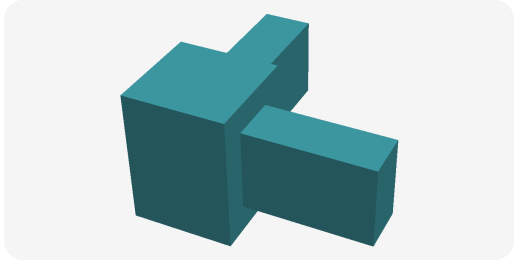
One unit per pane: `AFH` / `AFV` bars for the fixed pane, `ASH` / `ASV` for the sash, four `AC` corners each.

1. Assemble the spacer frame and fill its channels with silica gel.
2. Bond both 4 mm sheets to it with butyl glazing tape.
3. Peel the film off the inner faces only.

The sash unit is glazed in step 5, the fixed unit at install step 9. Bench window panes: fixed 498 × 228, sash 406 × 138 mm.

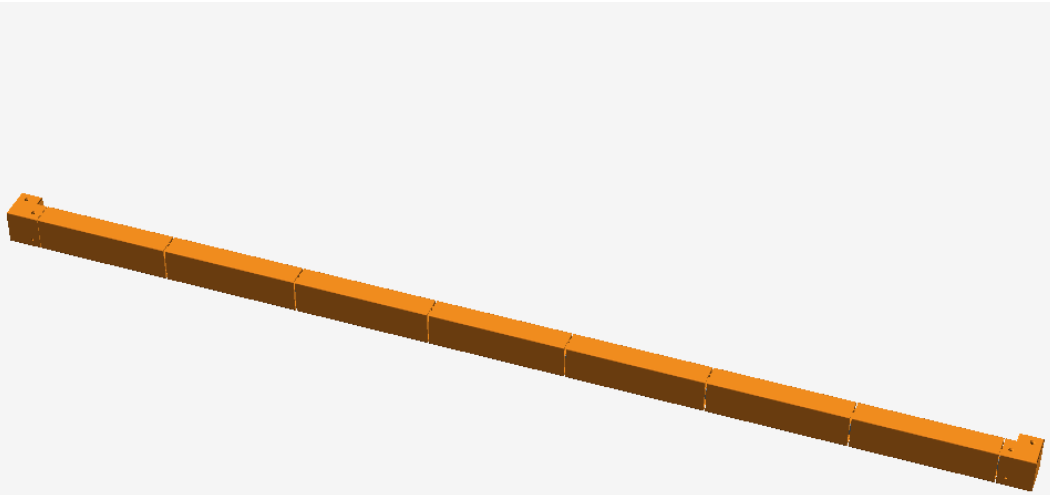


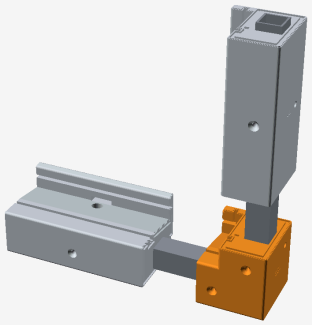
`AFH` spacer bar with its desiccant holes



`AC` corner, four per pane

Step 3a · Frame: the sill

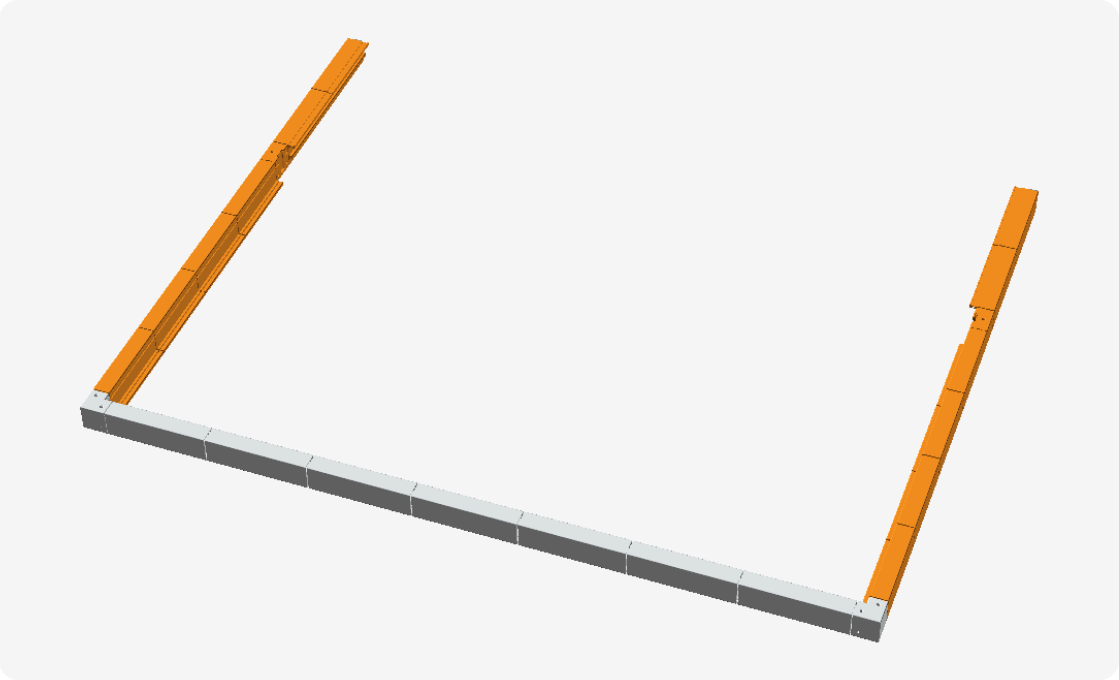


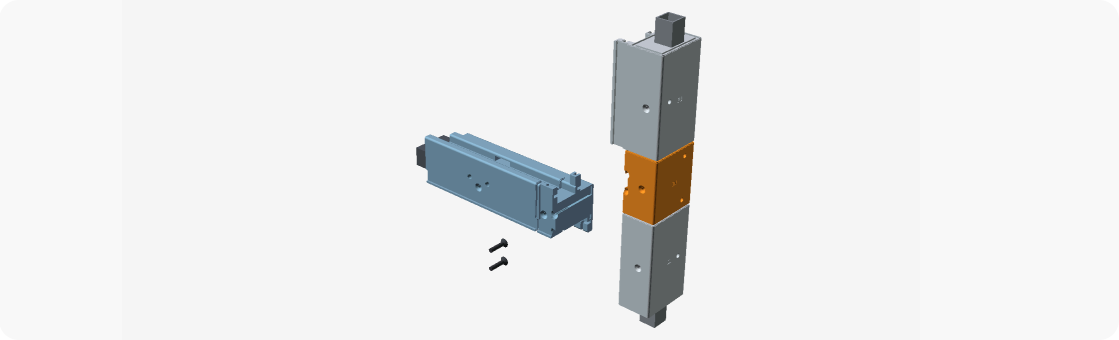


`CSL` with the sill and jamb tubes, `SW` and `JL` pulled back.

1. Fit corner `CSL` on the left end of the sill tube (as seen from outside), tube end 1.5 mm off the socket floor, and drive its sill-leg tube screw.
2. Thread on the sill segments: an `SW` (weep) first, then the 5 `S` segments, then the other `SW`. Fit `CSR` on the right end.
3. Close the joints one by one from `CSL`, following the joint rules, driving each segment's screw as you go. `CSR`'s screw goes in last.

Step 3b · Frame: jambs and nodes

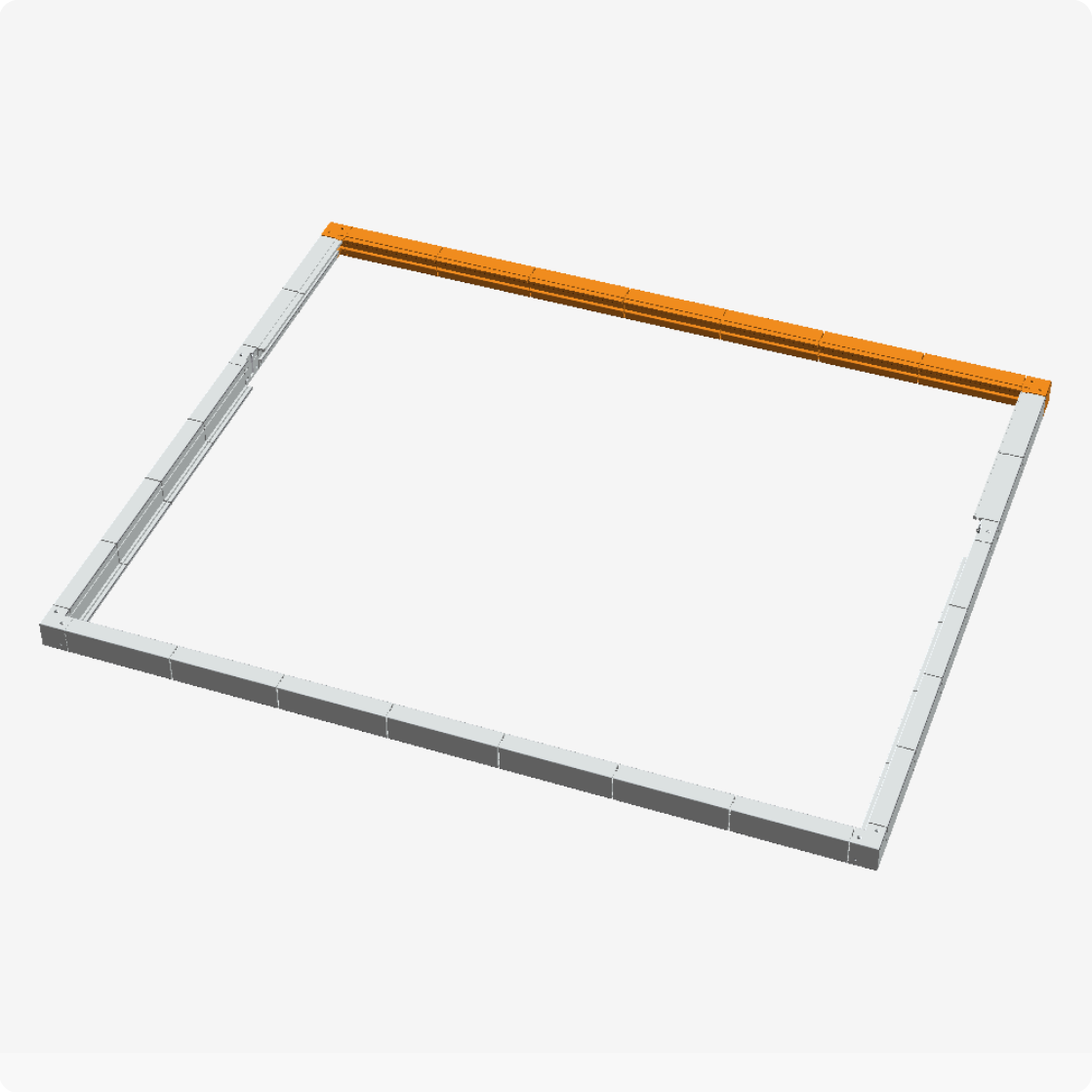




`JN` between `JL` and `JU`; the transom shoe comes later (install step 6).

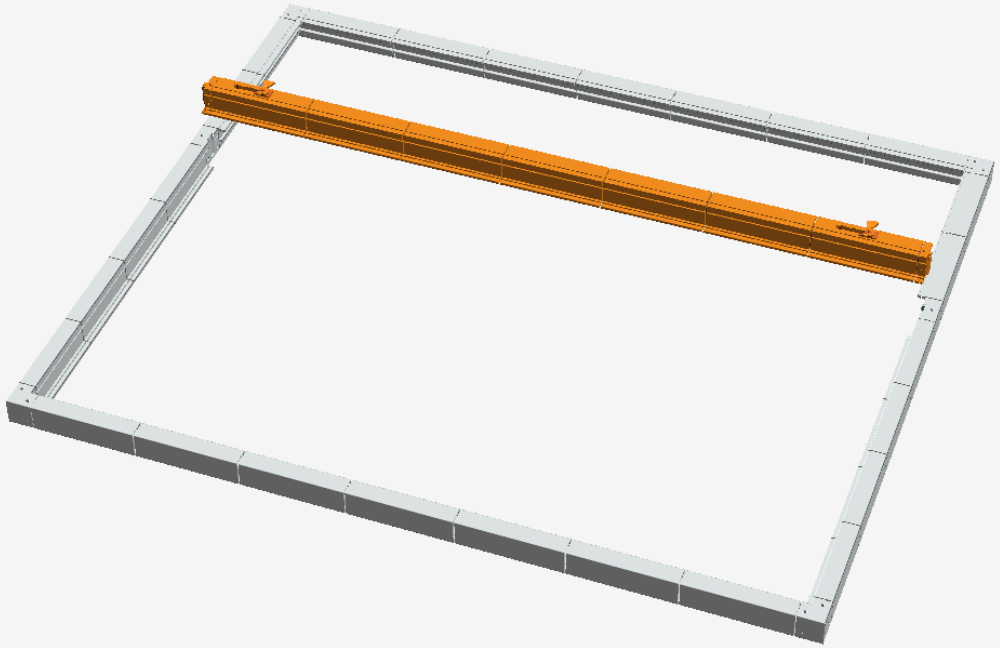
1. On each jamb tube thread the `JL` segments (4 per side), one `JN` node, then the `JU` segments (2 per side). With V-slot cores each jamb is two tubes (see the V-slot slide).
2. Push the tube into its sill corner's socket, tube end 1.5 mm off the socket floor.
3. Drive the sill corner's jamb-leg tube screw (`CSL` left, `CSR` right).
4. Close the joints from the bottom up, driving every `JL` and `JU` tube screw and the `JN` node's (inside face, centre of the node).

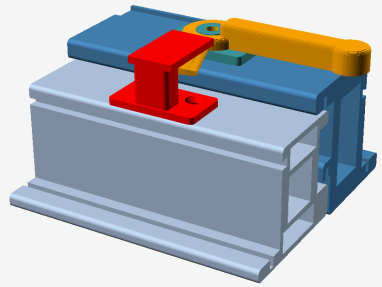
Step 3c · Frame: head, square, shims out



1. Assemble `CH`, the 7 `H` segments and `CH` on the head tube in the same way.
2. Butter the jamb-leg faces of both `CH` corners and drop the head onto the jamb tubes, closing the top jamb joints onto their shims.
3. Check the **diagonals are equal within 2 mm**, then drive the `CH` jamb-leg screws.
4. Pull out any shims still in, and fill and tool their voids.

Step 4 · Transom: cams and a dry fit



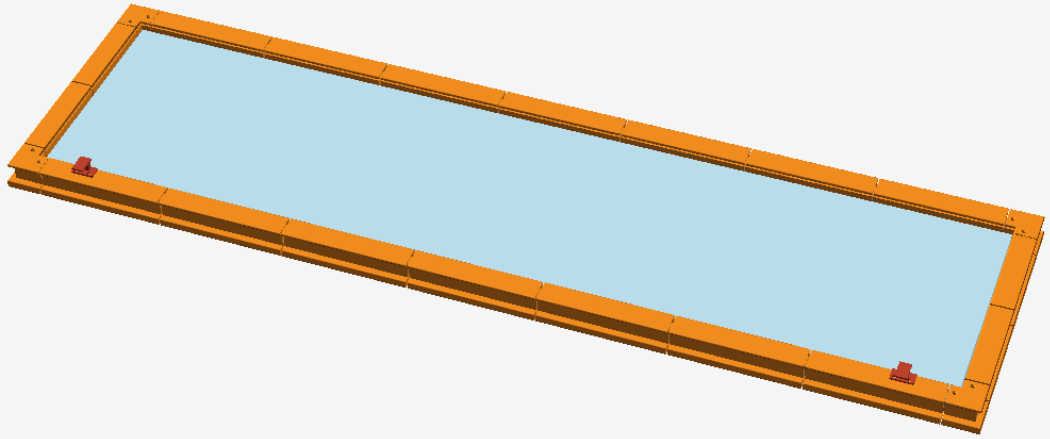


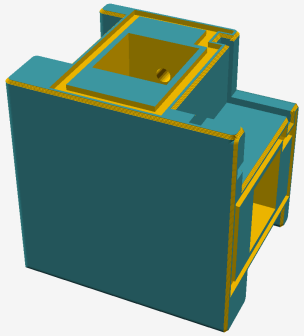
Cam base and handle on a `TE`; the keeper is fitted to the sash in step 5b.

The transom is only **dry-fitted** here: its length must match the node spacing of the installed frame, so it is sealed and screwed at install step 6.

1. **Cam bases:** screw the two `CMB` to the `TE` inserts, two M4 × 10 countersunk screws each.
2. **Handles:** fit each `CMH` over its pivot boss with an M4 × 8 pan screw and a 20 mm penny washer.
3. **Dry fit:** thread the `T` and `TE` segments on the steel tube, a `TE` at each end. Fit shoe `TSL` on the left end and `TSR` on the right (as seen from outside), each with a cord ring in its joint with the `TE`. Check every joint closes onto its shims; leave them unsealed and unscrewed.

Step 5a · Sash: assemble and glaze



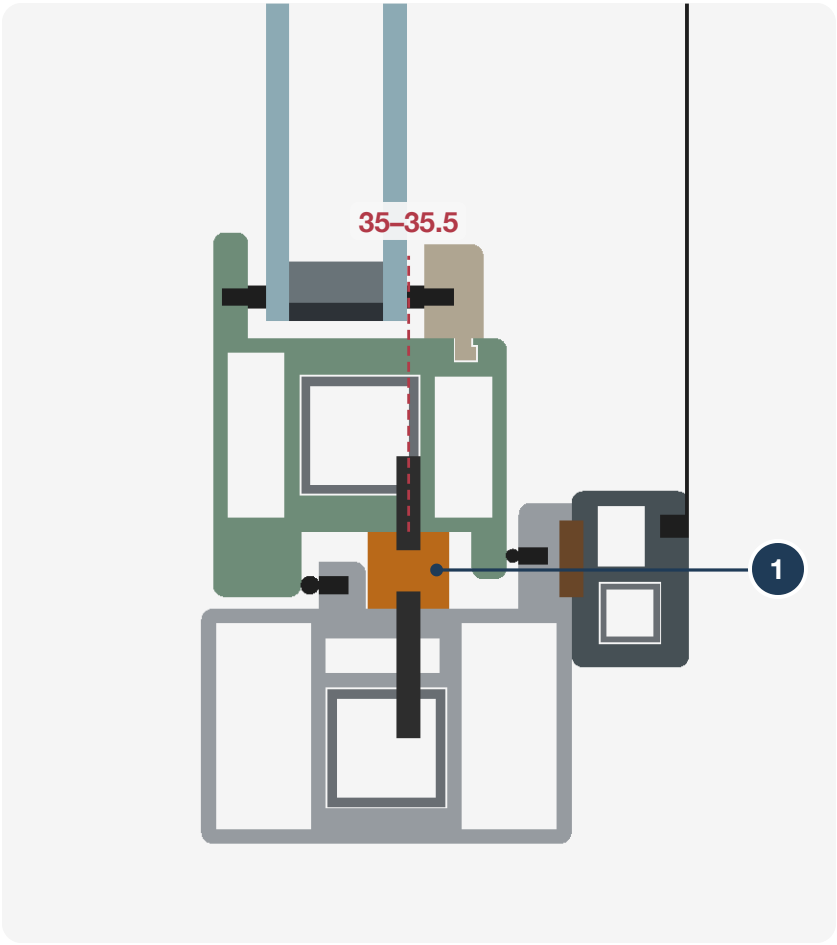


`CS` sash corner: both sash tubes end in its 21 mm sockets.

1. **Assemble:** fit `SH` (top), `SB` + `SBE` (bottom, `SBE` at the ends) and `SV` (sides) on their 20 × 20 tubes with four `CS` corners. Joint rules, 8g × 30 tube screws.
2. **Outer gasket:** fit the outer wedge gasket into the sash stop kerf all round.
3. **Blocks:** place four `GF`: two on the bottom rail at the quarter points of the glass width, one at mid-height on each side.
4. **Glass:** fit the IGU (or the acrylic unit).
5. **Beads:** snap in the `BSH` and `BSV` beads with the inner wedge gasket.

Step 5b · Keepers and friction stays (sash side)

1. **Keepers:** screw each `KP` to the **outer** boss pair of its `SBE` (nearer the sash corner) with two M4 × 10 countersunk screws.
2. **Stay arms:** screw each stay's sash arm to a sash side. The tracks go on the jambs at install step 8.
   - Stay size: stack ≤ 13 mm, track and arm ≤ 14 mm wide, rated ≥ 15.7 kg.
   - Along the sash: follow the stay's datasheet.
   - Across the sash side: centre each arm **33–33.5 mm in from the sash's outside face** (35–35.5 mm from the frame's outside face once hung).
   - Screws: 8g × 16 stainless countersunk, in 3.2 mm pilots through the shell into the tube, using the stay as the template (V-slot cores: 3.5 mm pilots, see the V-slot slide).



Plan section: the folded stay (1) between the jamb and the sash side.

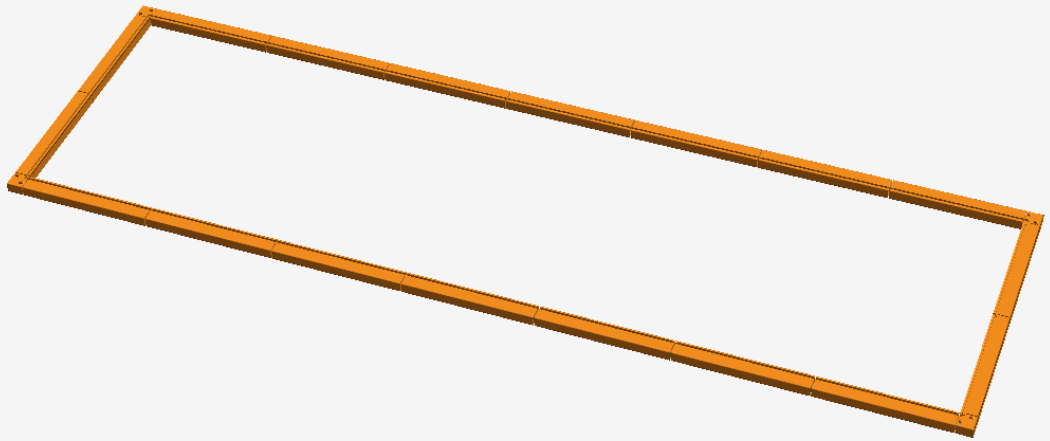
V-slot cores (optional)

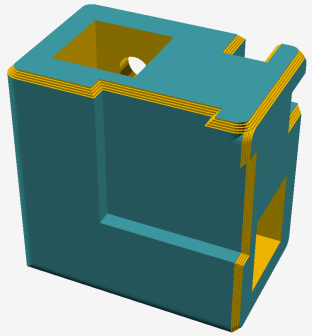
Only if the params file has `"core": "vslot"` (build guide section 8). The frame and sash cores are V-slot 2020; the transom and mullions are 2020 pairs. Every other step still applies: only the tubes and their screws change.

1. **Check the extrusion** first (test-log row 10): groove mouth 9.0–9.4 mm, slot depth 5.9–6.3 mm.
2. **Cut** to the V-slot cut plan in `bom/<name>.md`.
3. **Jambs are two tubes:** the lower tube runs from the sill corner into `JN`, the upper tube from `JN` to the head corner. The head and upper jambs sit **3 mm further out**.
4. **M5 countersunk screws** (lengths from the BOM) into slide-in T-nuts, with no drilling. Hang each nut with the **carrier**, a ~150 mm V-slot offcut:
   - put the nut in the offcut's slot and slide the offcut into the part's empty pocket until the nut is under the screw hole;
   - drive the screw until it catches the nut and the nut is snug against the offcut's lips;
   - back the screw off a quarter turn and slide the offcut out. The nut's neck now sits level with the pocket wall.
5. **Joins** go inside the segment the BOM names, with the piece end a quarter of the way along it.
6. **Transom pair:** the outer bar is trapped by the shoe end walls; the room-side bar takes 8g × 20 self-tappers in **3.5 mm pilots**.
7. **Friction stays:** same positions, **3.5 mm pilots** instead of 3.2 mm.
8. **Standard tubes:** members the BOM lists as standard keep a standard tube.

! Tighten each T-nut screw snug, about 2 N·m: the head pulls the printed wall against the aluminium.

Step 6 · Screen





`CN` screen corner, epoxied on both butt faces.

1. **Rails:** thread the `NH` and `NV` segments on their 10 × 10 tubes (horizontal rails: 7 segments, on a tube; vertical rails: 2 segments, on a tube). Close the joints tight: no shims, no cord. Drive an 8g × 12 screw (3.2 mm pilot) into each segment with a tube.
2. **Corners:** glue each rail end into its `CN` with 5-minute epoxy on both butt faces, drive the `CN` tube screw in every leg that takes a tube, and check the diagonals before the epoxy sets.
3. **Mesh and strip:** roll in the mesh with spline. Stick magnetic strip A into the screen's back channel all round.
4. Leave the hinge leaves off: `HG` and `HS` are fitted and aligned in place at install step 12.

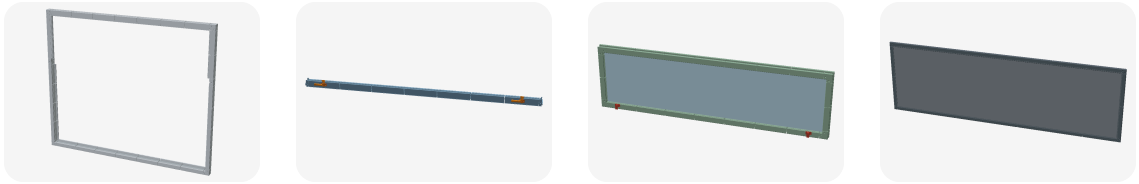
Before you start, and steps 1–2

Plan the day

- **Two people.** A dry day, with no rain forecast for 24 h.
- All four units built and checked; glass on site (1698 × 878 fixed IGU, about 30 kg).
- Fixings, packers, tape, backer rod, sealant and foam from the BOM.
- Work from a stable platform or ladder; gloves and eye protection when handling glass.

The 13 steps

1 remove · 2 measure · 3 sill tray · 4 set the frame · 5 fix it · 6 transom · 7 seals and magnet strip · 8 hang the sash · 9 glaze the fixed pane · 10 seal outside · 11 inside · 12 screen · 13 cams



Frame, transom, sash and screen: the four units you carry to the opening.

1 · Remove the old window

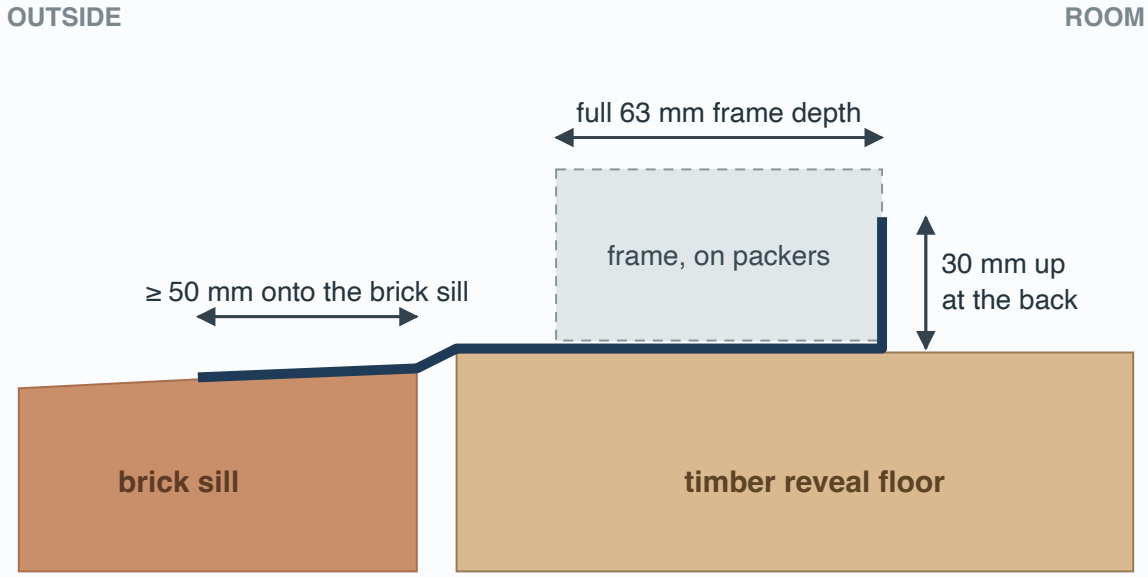
Unscrew the old frame from the timber reveal, cut the sealant to the brickwork and lift it out. **Keep the existing head flashing above the lintel intact.**

2 · Measure

- Width reveal to reveal and height head to brick sill, three points each; use the smallest.
- Timber reveal at least 63 mm deep.
- Put the numbers in your params file, regenerate and rebuild the BOM (Size it section). **Only print after this measurement.**

Step 3 · Sill tray

150 mm butyl flashing tape, ends turned up 100 mm.
With 100 mm tape: outer run first, inner run lapped 50 mm over it.



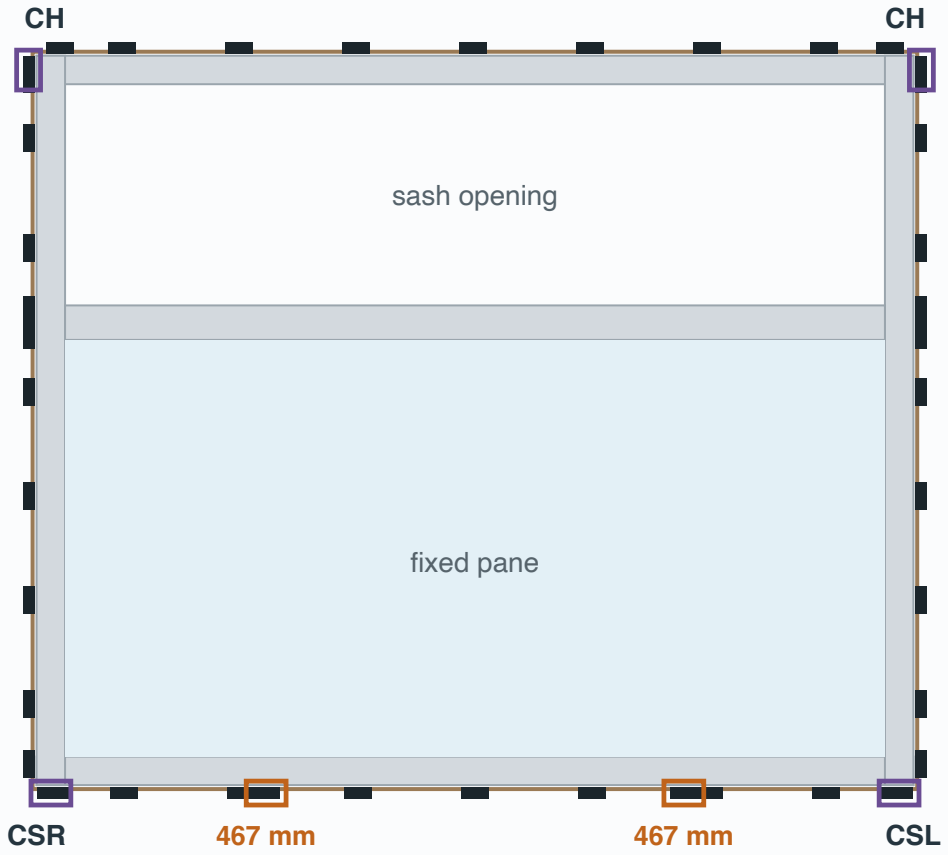
Schematic section through the sill, not to scale.

Lay 2.08 m of 150 mm butyl flashing tape across the opening floor:

- turned up **30 mm** at the back (room side);
- across the full **63 mm** frame depth;
- running out at least **50 mm** over the brick sill;
- turned up **100 mm** at each end.

With 100 mm tape, lay two runs: the outer run first, then the inner run lapped 50 mm over it, so the lap sheds water outward.

Step 4 · Set the frame on packers

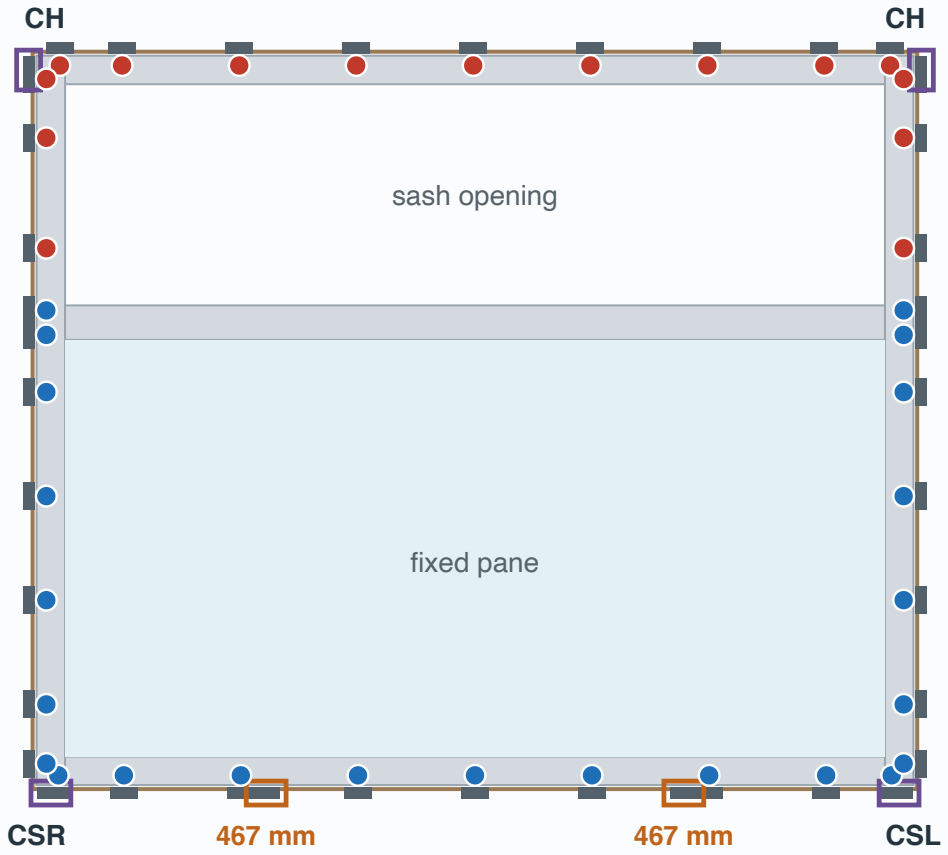


Seen from the room. Packers and fixings to scale in position, not in size.

1. Stand the frame on plastic packers under the two **sill corners** (purple) and under each `GW` **quarter point** (orange): 467 mm in from each outer frame edge on this window. Mark them on the sill. These two carry the fixed pane's weight straight into the wall.
2. Centre the frame, then plumb and square it: **diagonals within 2 mm**.
3. Wedge it with packers between the jambs and the reveal at the two **head corners**.
4. Tuck the head under the existing head flashing.

> The other black marks are the packers behind each fixing in step 5: 44 packer stacks in all.

Step 5 · Fix it



Seen from the room. Packers and fixings to scale in position, not in size.

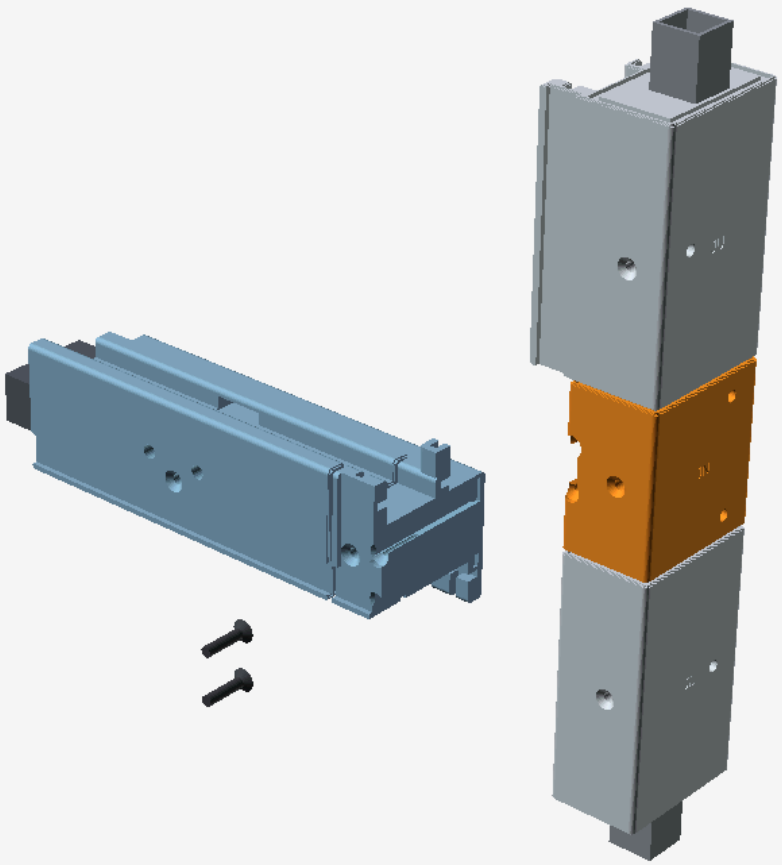
- Drive a **10g × 75** screw through every fixing hole into the reveal and studs (38 here), with a packer behind each, so the frame doesn't bow. Holes sit at each segment middle, both legs of every corner and two in each node.
- The node fixings carry the transom: pack them tight. The screws should hold, not crush.

Bed it: sill `S` / `SW`, both legs of `CSL` / `CSR`, lower jambs `JL` and both node fixings `JN` open into the drained pocket or outside the sash seal. Bed each screw in MS polymer and fill its counterbore flush.

Dry: head `H`, upper jambs `JU` and `CH` sit between the two seals.

! Drive every frame fixing before glazing, and before the stay tracks go on in step 8.

Step 6 · Assemble and fit the transom



1. Measure between the two node faces (with the dovetail slots) at transom height, outside and inside edges; take the smaller. Design figure: **1704 mm**.
2. On the bench, seal, close and screw the dry-fitted transom with the joint rules. Spread its shims (0.5–1.5 mm each) so the shoes' outer faces end up **0.5 mm closer** than the measured distance. Over 2 mm to close up? Trim the tube first. Drive the tek screws.
3. Lay 3 mm foam cord in the groove across each shoe's outer end face; butter the rest of the face with MS polymer.
4. Lift the transom in from the room side and slide the shoe tongues down the node dovetails to the stop.
5. Drive the four **M4 × 16** countersunk lock screws down the dovetail flanks; their heads finish under the magnet strip.
6. Tool the squeezed-out sealant along each shoe/node seam.

Step 7 · Frame seals and magnet strip

EPDM bulb seals: two continuous loops

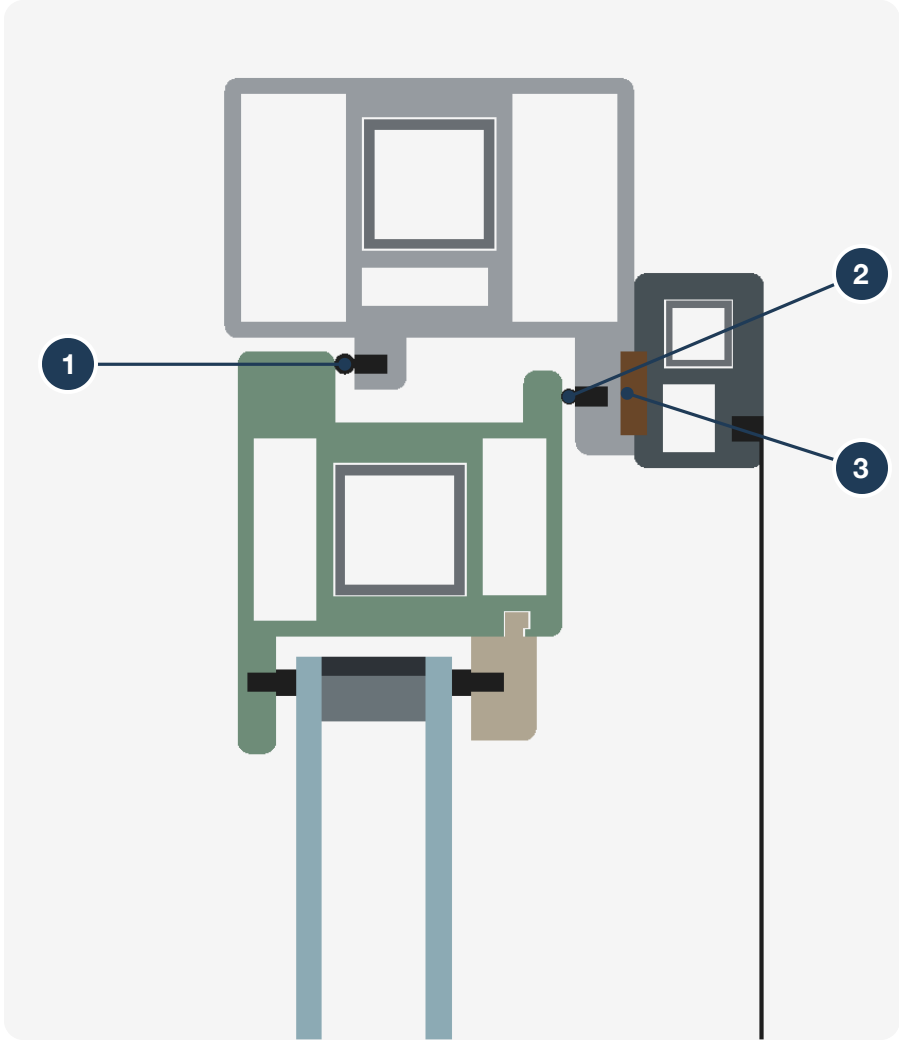
Outer seal in the fin kerf, inner seal in the inner-stop kerf; 8.92 m in all.

1. Start each loop at the middle of the head.
2. Run it across the head, down an upper jamb, through the shoe turn, along the transom, through the other shoe turn, up the other jamb and back.
3. Push it in **without stretching** it.
4. Cut the end square and butt it to the start with a drop of gel CA.

Magnetic strip B

Press it into the frame's magnet channel: along the **lower half of the transom's inside face**, through each shoe turn, up both upper jambs and across the head (4.53 m). It covers the lock-screw heads.

> **i** Strip B must be the matched partner of the screen's strip A, or a steel strip if A is a plain magnet.

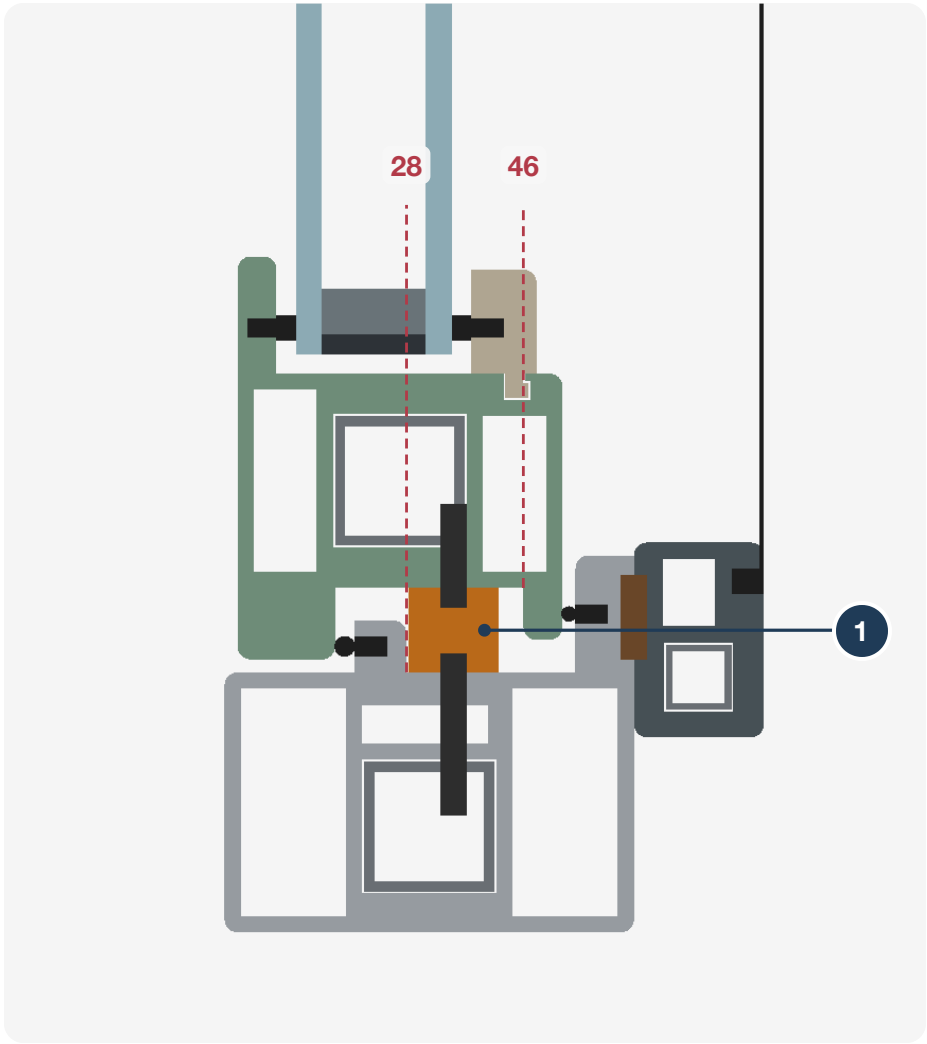


Head section: outer seal (1), inner seal (2), strip B (3).

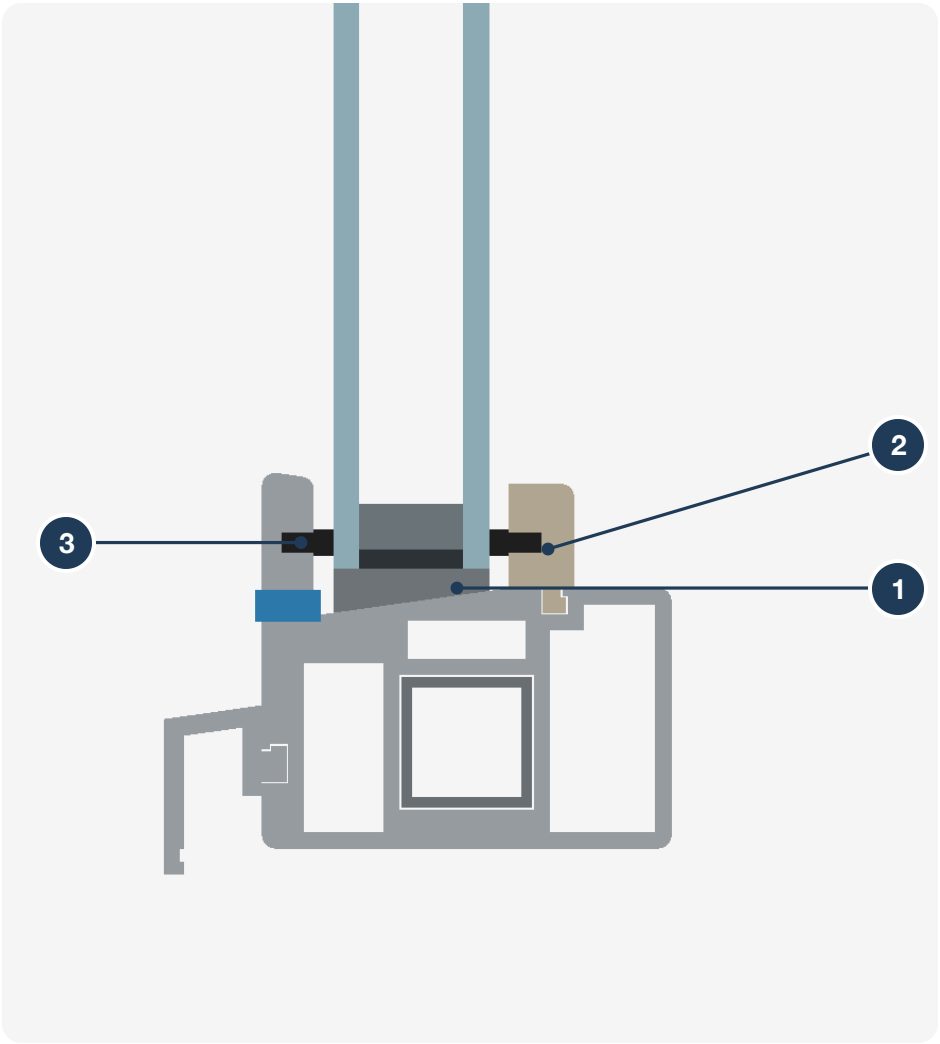
Step 8 · Hang the sash

Two people. All frame fixings are in, and the stays are already on the sash.

1. Hold the sash in the opening and place each stay's track against its upper jamb, at the height the stay's datasheet shows.
2. Centre each track **35–35.5 mm** in from the frame's outside face, level with its sash arm. Its body sits between the jamb fin top (28) and the sash lip (46); its screws hit the jamb tube (21.5–41.5 mm in; V-slot cores: see the V-slot slide) below the `JU` fixing counterbores (43–53 mm in).
3. Drill 3.2 mm pilots (3.5 mm with V-slot cores) through the track holes and screw the track on with **8g × 25** stainless countersunk screws.
4. Close the sash and check both seals touch evenly; adjust with the stay screws if needed.



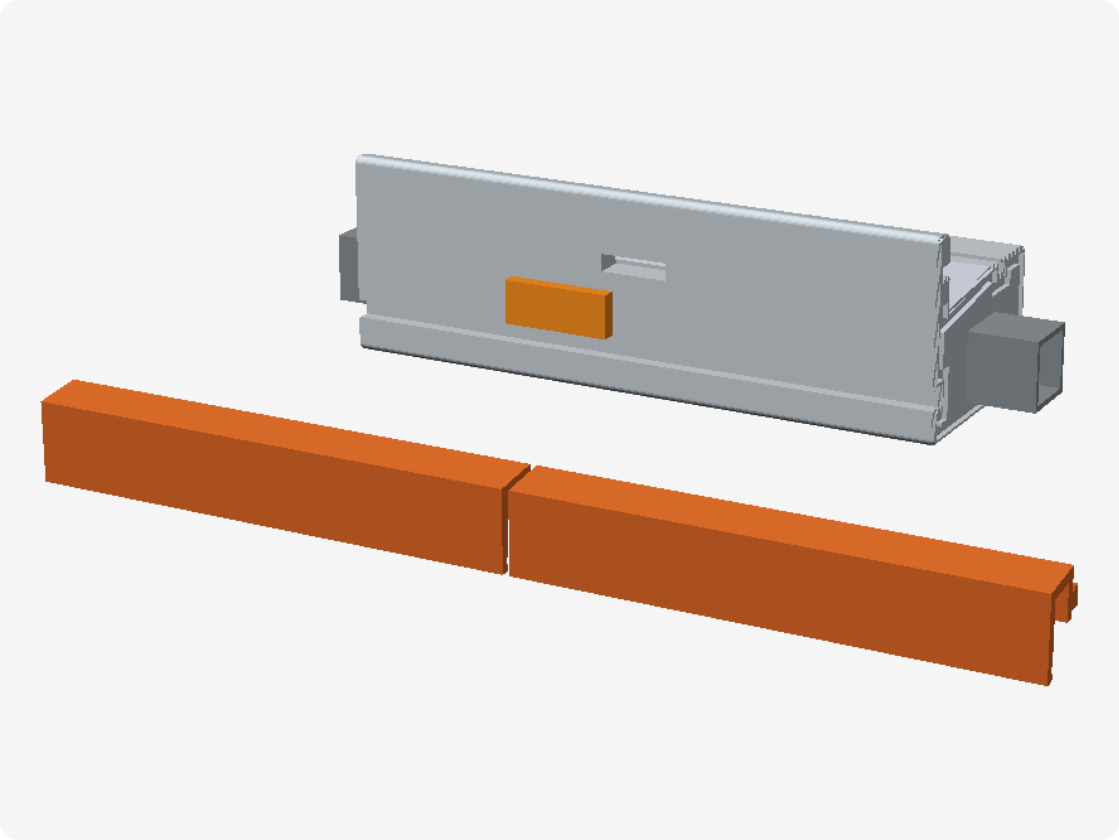
Step 9 · Glaze the fixed pane



1. Fit the outer wedge gasket into the stop kerfs (3).
2. Place two `GW` wedge blocks (1) on the sill floor at the quarter points, over the step-4 packers, and one `GF` locator at mid-height on each side.
3. Fit the glass, or the fixed acrylic unit.
4. Snap in the `BFH` / `BFV` beads (2) from the room side, with the inner wedge gasket.

! Lift the IGU with two people and suction lifters. Set it on the blocks; never on the bare sill.

Step 10 · Seal outside

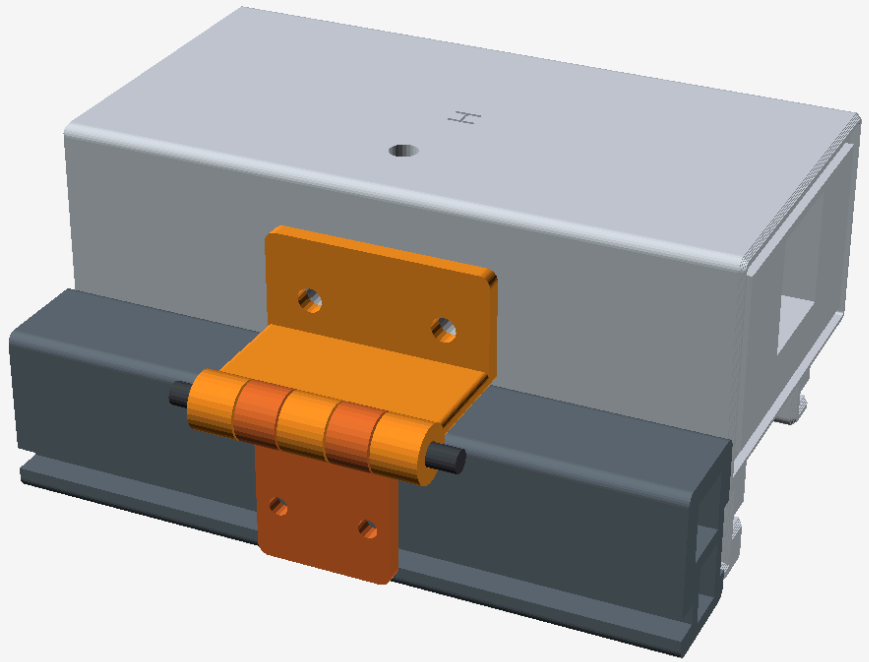


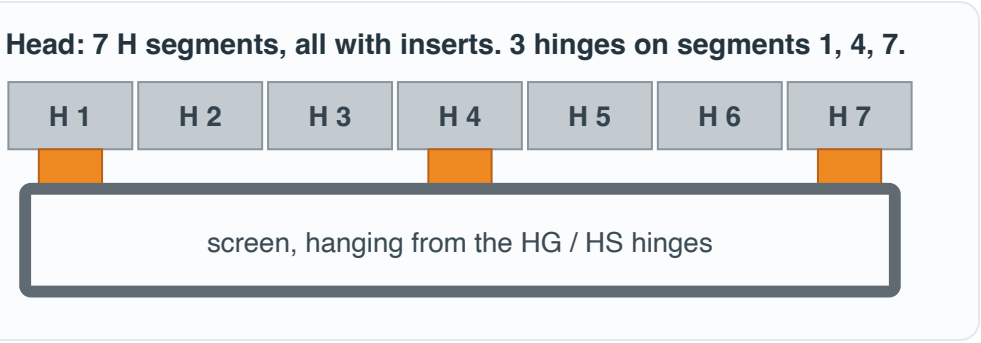
`SW` segment from outside: weep cap and two nose pieces pulled off.

1. Push backer rod into the gap between the frame and the brickwork (6.54 m), gun MS polymer over it and tool it smooth.
2. Clip on the sill-nose pieces `NS` with 1 mm gaps between them. The end pieces stop 17 mm from each frame edge. Tool a small fillet of MS polymer from each nose end to the perimeter seal.
3. Push in the four weep caps `WC`, open end facing down: two on the `SW` sill weeps and two on the `SBE` sash weeps (open the sash to reach them). The `TE` fin weeps get no cap.

! Never seal over weep holes or the nose drip.

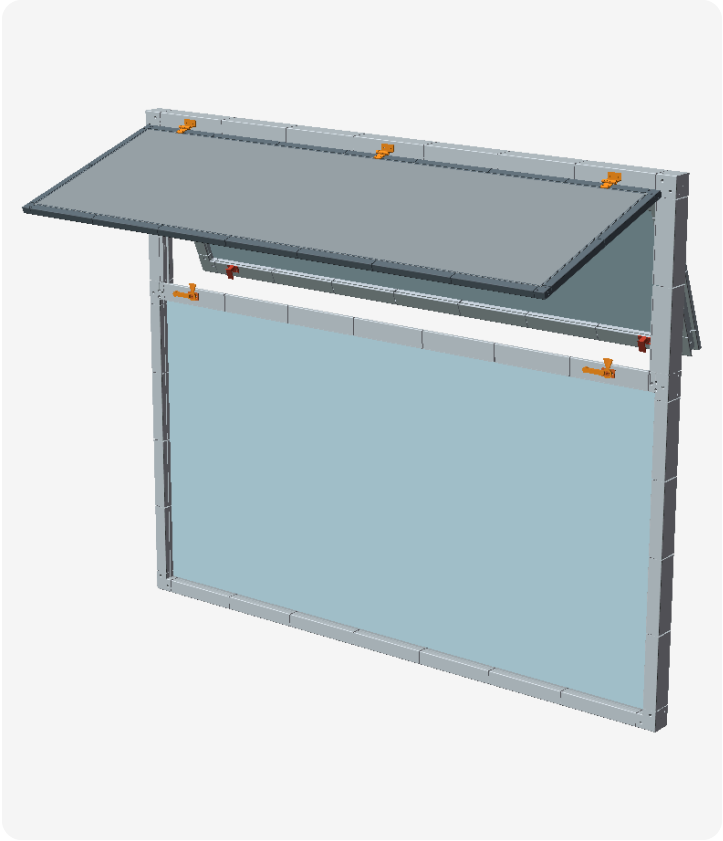
Steps 11–13 · Inside, screen, cams





11. **Inside:** fill the gap to the reveal with low-expansion foam, trim it and refit the architraves.
12. **Screen:** screw the `HG` leaves to the head inserts (two M4 × 10 pan or button heads each). Pin an `HS` into each with an M4 × 50 bolt and nyloc, snug but free to turn. Hold the screen closed on its magnets with an even **10 mm overlap** all round, swing the `HS` leaves onto its top rail and screw them with 4g × 12 screws (2 mm pilots). Check the strip closes along all four sides.
13. **Cams:** with the sash closed, turn each lever 60° counter-clockwise. The sash pulls in about 3 mm and both seals compress evenly.

Operating the window



Screen lifted, sash pushed out.

Opening the sash

1. Lift the insect screen by its bottom rail. It is hinged along the top and swings up towards the room.
2. Turn both cam levers (`CMH`) back, clockwise as seen from the room, about 60°, until each cam flag is clear of its keeper (`KP`) on the sash.
3. Push the sash out by its bottom rail. The friction stays hold it wherever you stop. Don't force it past the stays' end stop.
4. Lower the screen. It closes itself on its magnets, so the sash can stay open with the screen shut.

Closing and latching

1. Lift the screen.
2. Pull the sash in by its bottom rail until it sits evenly on both seals all round.
3. Turn each cam lever about 60° counter-clockwise. The flag rides up the keeper and pulls the sash in about 3 mm, so both seals compress evenly. Latch both cams, not just one.
4. Lower the screen.

> **i** The two cams are latches that hold the sash on its seals. They are not a keyed security lock.

A window split into bays by mullions has a sash, two cams and a screen in every bay. Open, close and latch each one on its own, as above.

Screen, ventilation and weather

Screen

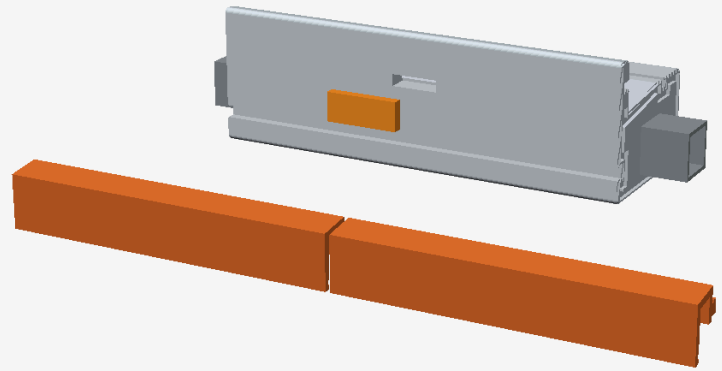
- The screen hangs on printed pin hinges (`HG` on the head, `HS` on the screen, an M4 × 50 bolt and nyloc as the pin): three on a wide window, two on a narrow one. Gravity and a matched pair of magnetic strips (A on the screen, B on the frame) close it.
- Open it by the bottom rail only, and let it close gently. Don't hang anything on it or lean on it.
- If light shows at the bottom corners, where the screen's magnet line leaves the transom, stick a thin foam pad there (test-log row 7).
- If it won't close by itself, the hinge pins are too tight: back off the nyloc nuts until the knuckles turn freely.
- To take the screen off, remove the three hinge bolts. To re-mesh it, pull the old 4 mm spline, roll in new mesh and spline, and trim.

Ventilation and weather

- A top-hung awning sheds light rain when it is open a little: water runs down the outside of the glass and off the sash. Keep the opening small in rain.
- Close and latch the sash in strong wind, storms or driving rain, and when nobody is home. The friction stays are sized for the sash's weight, not for gusts.
- Water that gets past the outer weather seal drains out of the chamber between the two seals through the weep slots in the transom fin (`TE` segments). Water in the glazing pockets drains through the sill weeps (`SW`) and sash weeps (`SBE`), each behind a flap cap (`WC`). These openings are part of the design: never seal, paint or block them.

Cleaning

- **Glass and frame faces:** warm water with a little dish detergent, a soft cloth or sponge, and a squeegee. No solvents (acetone, thinners, methylated spirits on painted parts), no abrasive pads or powders: they attack printed plastics and paint.
- **Acrylic glazing:** use a plastic-safe cleaner and a microfibre cloth. Acrylic scratches easily, and ammonia or solvent cleaners craze it.
- **Replacement acrylic:** cut new panels to the sizes in `bom/<name>-glass.csv`, in the same thickness as the window was built for (the glass list names it, e.g. "2 x 3 mm clear acrylic, 12 mm gap"). A different thickness needs re-printed beads and spacers.
- **Outside faces:** clean the fixed pane and the outside of the sash from outside, from the ground or a safe ladder or platform. Don't lean out through the open sash.
- **Seal kerfs and seals:** vacuum or brush out grit, and wipe the EPDM with water. Keep oils and petroleum products off EPDM.
- **Weeps:** pull each weep cap (`WC`), clear the slot with a pipe cleaner, and push the cap back with its open end facing down. Brush out the transom fin slots with the sash open.
- **Screen mesh:** take the screen off and wash it with soapy water and a soft brush, or vacuum it with a brush head.



Weep cap `WC` on an `SW` sill segment: pull, clear the slot, push back open end down.

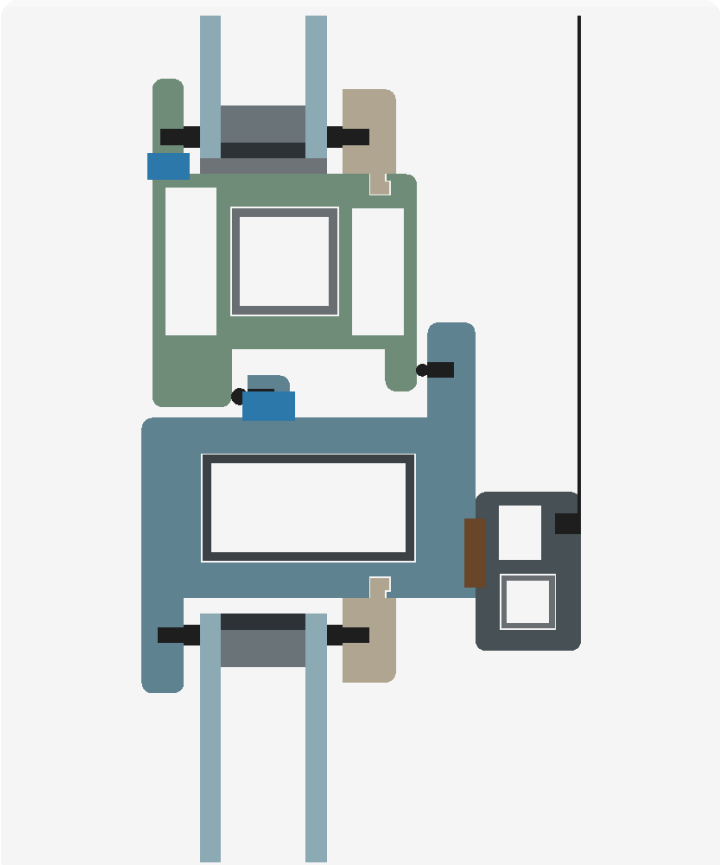
Maintenance schedule

When	Check and do
Monthly, and after storms	Weeps (`SW`, `SBE` caps, `TE` fin slots) and the sill-nose drip are clear; no water inside; screen closes on its magnets
Every season	Both EPDM seal loops are seated in their kerfs and their glued joints are closed; glazing wedge gaskets in place; mesh intact
Every 6 months	Friction stays: wipe clean, lubricate the pivots and slides as the stay maker specifies, and check their screws (8g × 25 in the tracks, 8g × 16 in the arms) are snug without over-tightening into the plastic. Cam locks: even pull-in, no cracks at the keeper post or flag, pivot screw snug
Yearly	Sealant at the segment joints (inside and outside faces), the perimeter seal to the brickwork and the sill-nose end fillets: cut out and re-fill any torn or loose bead with ±50 % MS polymer. Printed parts: look for chalking, cracks or creep, mostly on sunny faces. Hinge pins snug but turning
Every few years	Replace EPDM seals that have taken a permanent set or cracked, magnetic strips that no longer pull, and paint on painted (tier 1) parts once it chalks

ⓘ In the first year, check the joint sealant after the first hot spell and again after the first cold snap. Each 1 mm joint gap cycles between about 0 mm (hot) and 1.3 mm (cold), and that is when a weak bead tears.

Seasonal care

- **Winter:** condensation on the inside of the glass and frame is room moisture. Wipe it off and ventilate (crack the sash with the screen shut). Keep the weeps clear so the pockets can drain.
- **Fogging between panes:** in an IGU this means its edge seal has failed, and the unit must be replaced. In an acrylic sandwich the desiccant is saturated: open the unit, dry it and refill the silica gel (Troubleshooting).
- **Summer:** printed parts soften if the surface gets hotter than the material tier allows. Watch sunny and dark faces for distortion in the first summer. Keep light colours, and don't stick dark films or stickers on the frame.
- **Before storm season:** check the seals and sealant, and clear the weeps.



Weeps (blue) in the `TE` fin and `SBE` rail keep the pockets drained.

Troubleshooting (1/2)

Symptom	Likely cause	What to do
Draft at a segment joint	Sealant torn, often at the opening-side edge or in the band outside the cord ring	Cut out the loose sealant, clean, re-fill with ±50 % MS polymer and tool it, ideally at 15–25 °C
Draft round the sash	A seal loop out of its kerf, an open loop joint, or the cams not pulling in evenly	Re-seat the loop without stretching it, re-glue the joint with gel CA, latch both cams
Water inside	Blocked weeps, a torn joint or perimeter seal, or the head flashing not lapping the head	Clear the weeps first. Then hose-test from outside (test-log row 2) to find the leak, and re-seal
Condensation between acrylic panes	Desiccant saturated, or the butyl tape leaking	Unclip the beads, lift the unit out, open it, dry it, refill the silica gel and re-tape
Fog between IGU panes	Failed IGU edge seal	Replace the IGU. The size is in `bom/<name>-glass.csv`
Cam won't pull the sash in	Keeper loose or on the wrong boss pair, a seal out of its kerf, or the sash has dropped	Keeper screws (M4 × 10) on the outer boss pair of its **SBE**; re-seat the seals; check the stays and their screws

Troubleshooting (2/2)

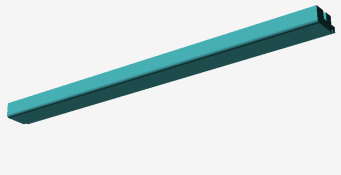
Symptom	Likely cause	What to do
Cam lever loose or wobbly	Pivot screw loose, or a worn handle	Snug the M4 × 8 pivot screw and its penny washer; re-print `CMH`
Gaps along the screen	Magnetic strip unstuck or not a matched pair, or the screen hanging crooked	Re-stick the strip, add foam pads at the bottom corners, re-align the hinge leaves
Sash sags, rubs or won't close evenly	Stay screws loose, a stay under-rated for the sash, or a sash corner creeping	Tighten the stay screws; check the stay rating is at least the sash weight; inspect the `CS` corners
Printed part cracked, warped or creeping	Material tier too low for the exposure, a dark colour, or a solvent	Re-print the part by its ID in the right tier and a light colour; paint tier-1 PETG
Glass rattles	Bead out of its keyway, or a wedge gasket missing	Snap the bead back in and refit the gasket

Safety

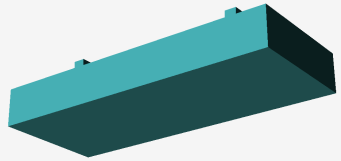
- **Glass:** order glass toughened wherever AS 1288 requires it; your glazier will know which panes. If a pane breaks, keep people away, wear gloves and eye protection, and tape or board the opening.
- **Weight:** the home window's fixed IGU weighs about 30 kg and the sash about 16 kg. Lift glass with two people and suction lifters.
- **Children and falls:** fit a window opening restrictor where the NCC requires one, for example on bedroom windows where the floor is 2 m or more above the ground below. Check your local requirements. The insect screen is not a barrier: never rely on it to stop a fall.
- **Not an emergency exit:** don't rely on the awning opening as an escape route.
- **Never lean on, hang from or put weight on an open sash,** and don't let children swing it.
- **Pinch points:** keep fingers clear of the sash edge and the cam levers when closing.
- **Working at height:** clean and maintain from the ground or a stable ladder or platform, never by leaning out of the window.
- **Not certified:** OpenWindow has no AS 2047 window rating and is not designed for bushfire-prone (BAL-rated) sites.

Spare parts and repairs

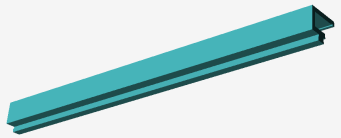
- **Re-print any part by its ID** from `cad/stl/<name>/<ID>.stl`: exterior parts in your tier's material, interior parts in PETG (`bom/<name>-parts.csv` says which). If the STLs are lost, regenerate them from the params file: `python3 tools/generate.py params/<name>.json`, then `cad/box.sh render <name>` (or export each `cad/generated/parts/<ID>.scad` in your own OpenSCAD).
- **Swap in place:** beads (`BFH`, `BFV`, `BSH`, `BSV`) unclip; weep caps (`WC`), sill-nose pieces (`NS`, re-tool the end fillets), cam handles and bases (`CMH`, `CMB`), keepers (`KP`), hinge leaves (`HG`, `HS`) and the whole screen unscrew.
- **Workshop jobs:** segments, corners, nodes and shoes are threaded on or screwed to the metal tubes. With V-slot cores, the tube screws are M5 into T-nuts, so they undo and go back in. A mullion (`MSL`, `ML`, `MST`, `MU`, `MSH`) slides out towards the room once its four lock screws are out and the seams are cut, but only after the parts around it come out. Take off the sashes and screens on both sides of it, peel off magnet strip B (it covers the lock screw heads) and the EPDM bulb loops that run through its channel and kerfs, then unclip the beads and lift out both fixed panes beside it: a lower mullion's stop legs sit outside the panes. A frame part means taking the window out, so a small crack in a frame segment is better bonded and sealed in place.
- **Off-the-shelf spares:** EPDM bulb kerf seal (3 mm kerf), EPDM glazing wedge gasket (3 mm), 4 mm closed-cell foam cord, 12.7 mm magnetic strip A and B, 4 mm screen spline, fibreglass mesh, ±50 % MS-polymer sealant, and the screws listed in `bom/<name>.md`.
- **Glass:** re-order from `bom/<name>-glass.csv` (or the glass table in `bom/<name>.md`).



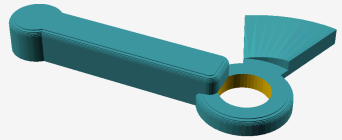
`BSH` Sash bead, horizontal



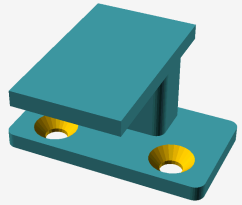
`WC` Weep cap



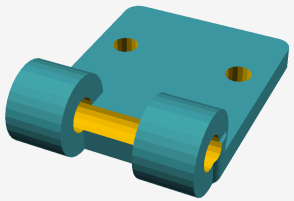
`NS` Sill nose



`CMH` Cam handle



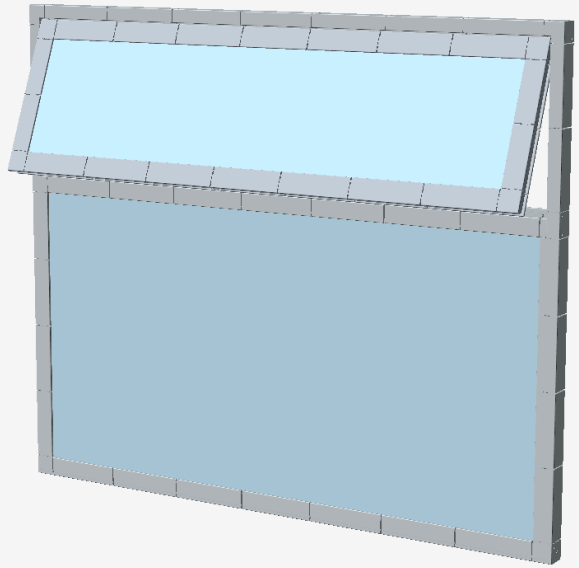
`KP` Cam keeper



`HS` Hinge leaf, screen

Specifications (1800 × 1500 home window)

Item	Value
Opening	1800 × 1500 mm, brick veneer, 8 mm gap each side
Frame	1784 × 1484 mm, 63 mm deep, 58 mm face
Layout	450 mm top awning over a fixed pane
Glazing	24 mm IGU 4/16/4: fixed 1698 × 878 mm, sash 1606 × 388 mm
Sash	About 15.7 kg on a 400 mm stainless friction-stay pair
Transom	40 × 20 × 1.6 steel tube: 6.2 mm at 0.7 kPa, limit span/175 = 9.5 mm
Seals	Two continuous EPDM bulb loops round the sash; wedge gaskets both sides of the glass
Printed parts	160 prints in 35 types: 16.36 kg exterior (tier 3) + 3.84 kg interior PETG
Glass thermal	About 2.7–3.0 W/m²K at the glass for double glazing (single glazing about 6): a design target, not a tested rating



Your own window's figures are in `bom/<name>.md` and in the lines `generate.py` prints.

Bench test protocol (1/3)

Bench window `params/bench-600.json` (acrylic glazing, tier 1), mounted in a timber test box (600 × 600 opening, 63 mm deep) and fixed exactly as in the install guide. Record each result in `docs/test-log.md`.

#	Test	Method	Pass criterion
1	**Fit**	Assemble from the printed parts with no filing	Every joint closes, diagonals within 2 mm, sash closes on both seals
2	**Water**	Garden hose, fan spray, 300 mm from the outside face, 15 min, sash closed and cammed. Aim at the opening-side edge of every segment joint, where the joint crosses the stops, fin and inner stop.	No water on the room side, including at the inside face of every joint; water leaves through the weeps
3	**Air**	Smoke pencil along both sash seal lines, the screen magnet line, and every segment joint, concentrating on its opening-side edge (stop, fin and inner-stop roots), with a fan blowing at the outside face	No visible smoke draw
4	**Heat/creep**	TC coupon painted matt black, 80 °C under a heat lamp (thermocouple on the face) for 24 h, with an M4 screw torqued to 1.5 N·m in one insert	No visible distortion; screw still holds 1.5 N·m

Bench test protocol (2/3)

#	Test	Method	Pass criterion
5	**Sash load**	Hang 2 × sash weight from the open sash's bottom rail for 1 h	No permanent set at the stays or the corners after unloading
6	**Cam**	Close and cam 50 times	Same compression on cycle 50 as on cycle 1, no cracking at keeper post or flag
7	**Screen corners**	Look for light at the screen's bottom corners (where its magnet strip leaves the transom)	No gap above 1 mm; if there is one, add foam pads and record where
8	**Mullion tees (windows with bays)**	Rows 2 and 3 again, aimed at the SN, TX and HN seams and at the shoe turns either side of each mullion	No water inside and no smoke draw at any mullion seam

Bench test protocol (3/3)

#	Test	Method	Pass criterion
9	**Mullion shoe pull-out (windows with bays)**	With the lock screws in, hook a luggage scale round the mullion next to one shoe and pull room-ward along the slot, by hand, to 200 N for 10 s. Do the shoe at the SN and the shoe at the TX. The mullion lock screws grip about half as much tongue as the jamb node's	No slip under 200 N, and no cracking at the shoe or the node
10	**V-slot profile (V-slot cores)**	Calipers on one piece: groove mouth width at the surface, and slot depth to the floor	Mouth 9.0–9.4 mm, depth 5.9–6.3 mm: the OpenBuilds profile that the screw positions assume
11	**V-slot stay screw pull-out (V-slot cores)**	On a JU segment on an offcut, drive an 8g × 25 screw through a 3.5 mm pilot at the stay line (35 mm in), then pull it straight out with a luggage scale to 300 N for 10 s. Repeat on the weakest case: an MU segment on two side-by-side offcuts, 8g × 16 in a 3.5 mm pilot at 35 mm in, to 300 N	No pull-out, and no cracking at the extrusion or the print
12	**T-nut hold (V-slot cores)**	A segment on an offcut with its M5 screw and T-nut snug: push the segment along the tube with a luggage scale to 150 N for 10 s	No slip, and no cracking round the countersink

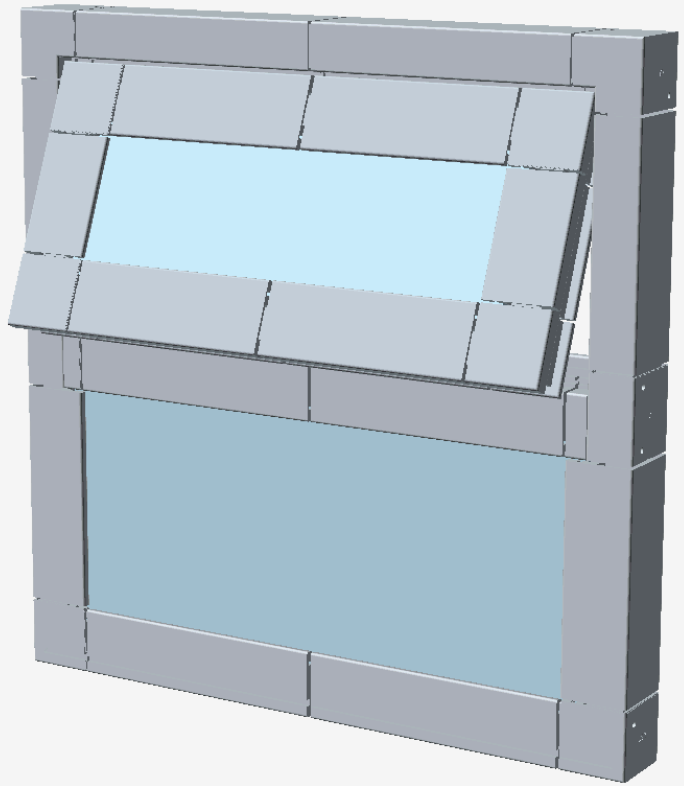
The joint sealant is assumed to take the movement: each 1 mm gap cycles between about 0 mm (hot) and 1.3 mm (cold). Repeat rows 2–3 after a hot day and a cold night, and record any joint whose sealant has torn or come away.

Rows 8–9 need a window with bays: `docs/test-log.md` gives a 2-bay bench window (`bench-1000-2bay`, 1000 × 600). Rows 10–12 are for V-slot cores: save `bench-600` as `params/bench-600-vslot.json` with `"name": "bench-600-vslot"` and `"core": "vslot"` added.

The gate before the full-size build

1. **Friction stay found:** a 400 mm pair rated ≥ 15.7 kg, stack ≤ 13 mm, width ≤ 14 mm. If none exists, the sash cavity or tube must change, and the frame, sash and corner parts be re-rendered.
2. **Bench rows 1–6 passed:** fit, water, air, heat/creep, sash load, cam. A window with bays also needs rows 8–9 (mullion tees, shoe pull-out) passed on a 2-bay bench window. V-slot cores run rows 1–6 on the V-slot bench window (`bench-600-vslot`) as well, since row 5 loads the sash through its M5 corners and V-slot stays, and also need rows 10–12 (profile, stay screw pull-out, T-nut hold).
3. **Opening measured,** params updated, `generate.py` clean (no `WARNING`), BOM rebuilt.
4. **Glass ordered** from `bom/<name>-glass.csv`, toughened where AS 1288 requires.
5. **Material tier** chosen for the wall's sun exposure, in a light colour.

✓ Then print the home window: 160 parts.



The 600 × 600 bench window: 98 prints, the same profiles and hardware.

A1 · Bench window bill of materials (1/3)

Group	Item	Qty	Unit
Fixings	Timber screw 10g x 75, stainless or galvanised (frame fixings)	20	pcs
Fixings	Self-tapping screw 8g x 30 stainless, countersunk (aluminium tube screws: frame, sash, corners, nodes)	32	pcs
Fixings	Tek screw 8g x 25, 410 stainless, countersunk (steel transom tube: T, TE, shoes)	4	pcs
Fixings	Self-tapping screw 8g x 12 stainless, countersunk (screen tube screws)	8	pcs
Fixings	M4 heat-set insert for a 5.6 x 8 mm hole	20	pcs
Fixings	M4 x 10 countersunk machine screw (cam bases CMB, keepers KP)	8	pcs
Fixings	M4 x 10 pan or button head machine screw (hinge frame leaves HG)	4	pcs
Fixings	M4 x 8 pan screw + 20 mm penny washer (cam pivots)	2	pcs
Fixings	M4 x 16 countersunk machine screw (transom lock screws)	4	pcs
Fixings	M4 x 50 bolt + nyloc nut (hinge pins)	2	pcs
Fixings	4g x 12 self-tapping screw (screen leaves HS)	4	pcs
Fixings	Plastic window packers, assorted 1-10 mm (one stack per position)	26	pcs
Hardware	Stainless friction stay pair, 200 mm, stack <= 13 mm, track/arm width <= 14 mm (check rating >= 1.6 kg)	1	pair

A1 · Bench window bill of materials (2/3)

Group	Item	Qty	Unit
Hardware	Friction-stay screw 8g x 25 stainless, countersunk (tracks to upper jambs; match the stay's holes)	8	pcs
Hardware	Friction-stay screw 8g x 16 stainless, countersunk (arms to sash sides; match the stay's holes)	6	pcs
Seals	EPDM bulb kerf seal, 3 mm kerf (outer + inner loop round the sash opening)	2.94	m
Seals	EPDM glazing wedge gasket, 3 mm	5.08	m
Seals	Closed-cell foam cord, 4 mm (one ring per segment joint)	4.9	m
Seals	Closed-cell foam cord, 3 mm (transom shoe seams)	0.14	m
Seals	MS-polymer sealant rated for +/-50% joint movement (ASTM C920 Class 50 or equal), 290 ml cartridge (joints, fixing bedding, outside perimeter)	2	pcs
Seals	Butyl flashing tape, 150 mm wide (sill tray; or two 100 mm runs lapped 50 mm)	0.88	m
Seals	Backer rod, 10 mm	2.34	m
Seals	Low-expansion window foam, can	1	pcs
Seals	Gel CA glue (EPDM loop joints)	1	pcs
Screen	Flexible magnetic strip A, 12.7 mm, adhesive-backed (screen back channel)	1.63	m
Screen	Flexible magnetic strip B, 12.7 mm, adhesive-backed, matched to A, or adhesive steel strip (frame magnet channel)	1.63	m

A1 · Bench window bill of materials (3/3)

Group	Item	Qty	Unit
Screen	Fibreglass insect mesh	0.17	m2
Screen	Screen spline, 4 mm	1.63	m
Screen	Two-part epoxy, 5-minute (screen corner joints)	1	pcs
Tube	Tube alu10 (total length; buy in 1 m/3 m stock)	0.97	m
Tube	Tube alu20 (total length; buy in 1 m/3 m stock)	3.21	m
Tube	Tube steel40x20 (total length; buy in 1 m/3 m stock)	0.5	m
Glazing	Acrylic 4 mm, 498 x 228 (fixed pane)	2	pcs
Glazing	Acrylic 4 mm, 406 x 138 (sash pane)	2	pcs
Glazing	Silica gel desiccant beads	100	g
Glazing	Butyl glazing tape, 3 mm (acrylic sandwich edge)	5.08	m
Filament	PETG (interior)	1.25	kg
Filament	PETG + exterior paint, or ASA (exterior, tier 1)	6.19	kg

A2 · Cut lists and glass: home and bench

home-1800x1500

Member	Core	mm	Qty
head	alu20	1721	1
sill	alu20	1721	1
jamb	alu20	1421	2
transom	steel40x20	1696	1
sash horizontal	alu20	1615	2
sash vertical	alu20	397	2
screen horizontal	alu10	1685	2
screen vertical	alu10	501	2

Pane	Glazing	W	H	Qty
fixed	igu	1698	878	1
sash	igu	1606	388	1

bench-600

Member	Core	mm	Qty
head	alu20	521	1
sill	alu20	521	1
jamb	alu20	521	2
transom	steel40x20	496	1
sash horizontal	alu20	415	2
sash vertical	alu20	147	2
screen horizontal	alu10	485	2

Pane	Glazing	W	H	Qty
fixed	acrylic	498	228	1
sash	acrylic	406	138	1

A3 · Key dimensions (`cad/dims.json`, mm)

Key		Value
`DEPTH`	frame depth	63
`FRAME_W`	frame face	58
`FRAME_BODY`	frame body	40
`TRANSOM_W`	transom width	70
`TRANSOM_HALF_BODY`	transom half body	17
`SASH_W`	sash width	51
`SASH_BODY`	sash body	33
`SASH_GAP`	sash gap (stay cavity)	13
`STAY_W`	max stay width	14
`SCREEN_W`	screen rail width	30

Key		Value
`SCREEN_DEPTH`	screen depth	20
`SCREEN_OVERLAP`	screen overlap	10
`GLASS_T`	glazing pocket	24
`GLASS_EDGE`	fixed-glass edge cover	43
`SASH_GLASS`	sash-glass edge cover	36
`BEAD_T`	bead thickness	10
`MAX_SEG`	max segment length	240
`JOINT_GAP`	joint gap	1
`CORD_D`	cord groove width	3
`MIN_SHELL`	min plastic round tube	6

Key		Value
`CHAMBER_WALL`	chamber wall	2.5
`SOCKET_FRAME`	frame tube socket	28
`SOCKET_SASH`	sash tube socket	21
`SOCKET_SCREEN`	screen tube socket	12
`SILL_SLOPE_DEG`	sill pocket slope (°)	8
`NOSE_PROJECTION`	sill-nose projection	15
`INSERT_M4_D`	insert hole Ø	5.6
`FIX_CB_D`	fixing counterbore Ø	10

A4 · Repository map

- `params/`: one JSON file per window
- `tools/window.py`: layout, segment splits, glass, tube cuts, checks
- `tools/generate.py`: part wrappers and CSVs in `cad/generated/`
- `tools/check_stl.py`, `stl.py`: STL checks; `test_*.py`: Python tests
- `cad/dims.json`: every shared dimension
- `cad/profile.scad`: every cross-section
- `cad/lib/`, `cad/parts/`: segments, corners, nodes, shoes, small parts
- `cad/tests/`: OpenSCAD assert tests and the review views
- `cad/box.sh`: render, test and png on the OpenSCAD machine
- `cad/stl/<name>/`: the printable STLs
- `cad/deck/`: views for this guide's illustrations
- `bom/build_bom.py` and `bom/<name>.*`: BOM, cut list, glass, part list
- `docs/`: design spec, plan, printing, build and install guides, test log, owner's manual, images
- `pdf/build_deck.py`: this guide, as `OpenWindow-Guide.pdf`

A5 · Rebuild everything, credits and licence

```
python3 tools/generate.py params/<name>.json
cad/box.sh render <name>
python3 bom/build_bom.py <name>
cd tools && python3 -m unittest discover -p 'test_*.py'
cad/box.sh test
python3 pdf/build_deck.py
```

- Deck illustrations: `python3 pdf/render_deck.py` (one line per image in `pdf/deck_renders.txt`)
- Check one slide: `python3 pdf/build_deck.py --slide N --png out.png`

Credits and licence

- OpenWindow v1: design, CAD, generator and guides by the OpenWindow project.
- Modelled in OpenSCAD (manifold backend); this guide is built with Python and headless Chrome.
- **Licence: to be chosen.** Until then, ask before redistributing.

> **i** Status: designed and checked in CAD and software only. No window has been printed or tested yet; the bench test is next.